

Interface descriptions

700026276-en

Die casting machine

ECOLINE 84

精锐

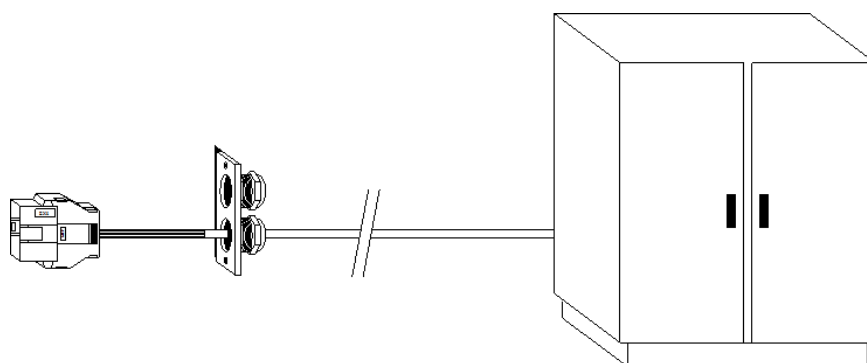


Table of contents

Buhler Interface for Die Spray Unit	GEB-19200-en-ECO.1D
Buhler Interface for Extraction-Robot	GEB-19100-en-ECO.1F
Buhler Interface for Ladling unit	GEB-19000-en-ECO.1E
Bühler interface for Vacuum	GEB-19600-en-ECO.1D
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Buhler Interface die spray unit (Ecoline)

Control:

Ecoline

Release

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Content

1	Introduction.....	4
1.1	Purpose	4
1.2	Abbreviations	4
1.3	Electrical Specification	4
1.4	Signal exchange.....	4
1.4.1	DCM Control --> Die Spray Unit.....	4
1.4.2	Die Spray Unit --> DCM Control.....	4
1.5	Safety	5
1.5.1	Signal monitoring implemented in DCM control	5
1.5.2	Personal safety.....	5
1.5.3	Responsibility	5
2	Connection-description.....	6
2.1	Cable	6
2.2	Plug	7
3	Signals: DCM Control --> Die spray unit	8
3.1	Operation mode automatic.....	8
3.2	Release of protective devices, signals 1 and 2.....	8
3.3	Die open	8
3.4	Release for entering into die area	8
3.5	Ejector in rear position	9
3.6	Command manual spraying	9
3.7	Command go to basic position.....	9
3.8	Die closed.....	9
3.9	Operation mode setting	10
3.10	Operation mode manual.....	10
4	Signals: Die spray unit --> DCM Control	11
4.1	Die spray unit is in operation mode A / SA	11
4.2	Die spray unit in basic position.....	11
4.3	Spraying completed (Impulse: min. 200ms).....	11
4.4	Die spray unit fault priority 2.....	11
4.5	Emergency-off	12
4.6	Die spray unit out of die area	12

1 Introduction

1.1 Purpose

The interface is suitable for:

- Spraying from the left*, the right* or the top*
- Spraying of the die
- Spraying of the ejector (with core spray program)

*) Has to be specified with your order

1.2 Abbreviations

S	Operation mode setting
M	Operation mode manual
SA	Operation mode semi-automatic
A	Operation mode automatic
DCM	Die casting machine
Q	Fault message can be acknowledged on the keyboard of the DCM control
NQ	Fault message cannot be acknowledged
Version 21	Indication of the second release of the interface with additional signals

1.3 Electrical Specification

Output signals of DCM control:

Potential free relays contactors max. 2A / max. AC 250V / DC 30V min. 10mA/5V.

Input signals of DCM control:

min. 10mA / 24V DC (0V till +5V are considered 0V and +15V till +30V are considered 24V.

Control voltage from DCM).

In the DCM control cabinet all signals are wired on plugs (see also signal allocation diagram of DCM).

1.4 Signal exchange

1.4.1 DCM Control ---> Die Spray Unit

All output signals are only present, when "die spray unit" is selected on screen.

1.4.2 Die Spray Unit ---> DCM Control

Except the signals "Die spray unit in basic position" and "Die spray unit out of die area", all input signals are only processed by the DCM control, when "die spray unit" is selected on screen.

1.5 Safety

1.5.1 Signal monitoring implemented in DCM control

Some signals are monitored for cyclic actuation, e.g. these signals must change their status at least once during one DCM cycle. Otherwise the installation changes to manual mode (end installation cycle) and a corresponding message will be displayed on the screen (see signal description).

1.5.2 Personal safety

According to safety regulations the operator must not be endangered by movements of the unit. (-- > place a fence). Apertures in the protecting system of the working place, such as doors in the fencing and the safety gate on the operator's side of the DCM, must be linked with the sprayer. To this end potential-free signals (2-channel) have been provided on the interface.

Signal "Release of protective devices" (for details see 3.2.):

The sprayer may enter the die area exclusively when these signals are present. If the signals drop off while the sprayer is operating within the die area the unit must stop immediately and the drive units must be switched off (regulation from EN 869). If the sprayer is placed on the left or right hand side of the DCM, the unit must be fenced in. The drive units have to be interlocked with the door of the fence.

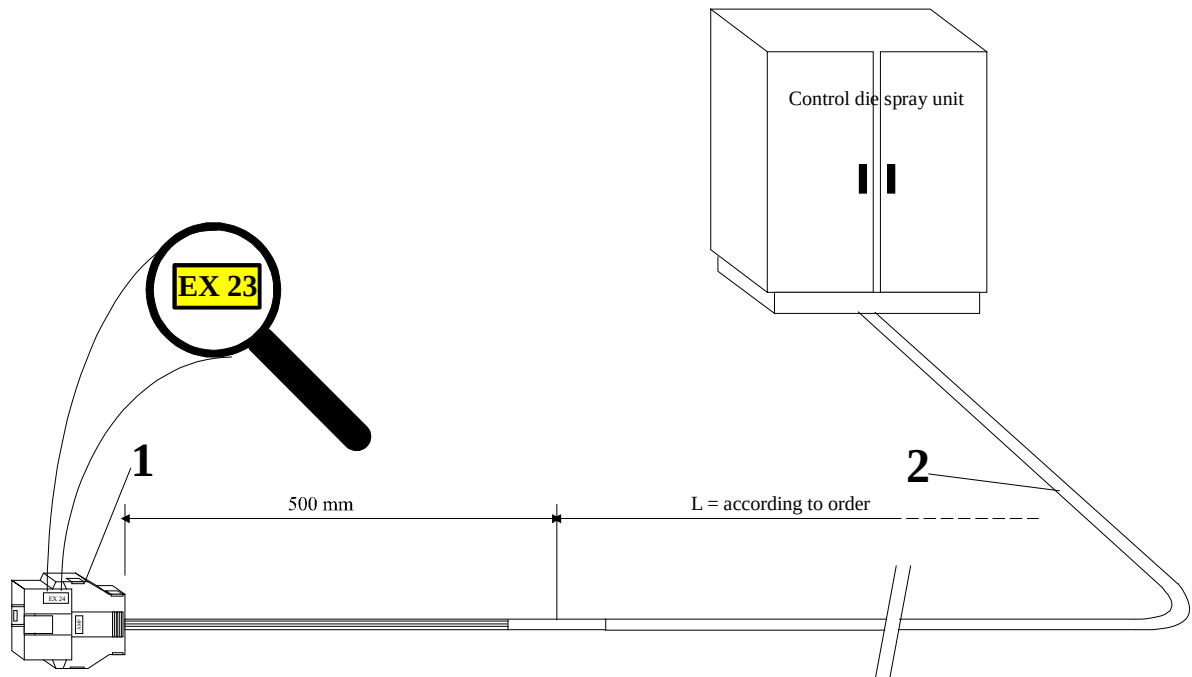
1.5.3 Responsibility

If the unit is furnished by the customer, he carries the responsibility for conformation of the unit to regulations specific the country (e.g. CE) and that it is linked to the installation in accordance.

Before setting into operation the effective location of the unit (to the left or right or on top) corresponds with the position as predetermined in the DCM (symbol of die spray unit on page "Information" = initial image). If this is not the case this must be altered immediately (contact customer service).

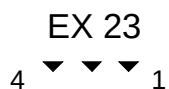
2 Connection-description

2.1 Cable



The interface cable (1+2) will be manufactured by the producer of the periphery gear. A plug package (1) can be ordered from Bühler AG.

2.2 Plug



- 1: 24 Volt from DCM
- 2: GND from DCM
- 3: SPIATA: Die spray unit is in operation mode A / SA
- 4: SPIGS: Die spray unit is in basic position
- 5: SPAFOB: Die spray unit out of die area
- 6:
- 7:
- 8: SPEND: Spraying completed
- 9:
- 10: SPST2: Die spray unit fault priority 2
- 11:
- 12: Emergency-off die spray unit → DCM, channel 1
- 13: Emergency-off message
- 14: Emergency-off die spray unit → DCM, channel 1
- 15: Emergency-off DCM → die spray unit, channel 1
- 16: Emergency-off DCM → die spray unit, channel 1
- 17: Release of protective devices, signal 1
- 18: Release of protective devices, signal 1
- 19: Release of protective devices, signal 2
- 20: Release of protective devices, signal 2
- 21: Emergency-off die spray unit → DCM, channel 2
- 22: Emergency-off die spray unit → DCM, channel 2
- 23: Emergency-off DCM → die spray unit, channel 2
- 24: Emergency-off DCM → die spray unit, channel 2
- 25: 24 Volt from spray unit
- 26: SPBAA: Operating mode automatic
- 27: SPFLSO1: Die open
- 28: SPFFOB: Release for entering into die area
- 29: SPAUSZ: Ejector in rear position
- 30: SPBZH: Command manual spraying
- 31: SPGSA: Command go to basic position
- 32: SPFLSG: Die closed or
SPBAE : Operation mode : setting
- 33: SPBAH ; Operation mode : manual
- 34:
- 35:
- 36:

3 Signals: DCM Control --> Die spray unit

3.1 Operation mode automatic

SPBAA	EX23:26
S,M	----
SA,A	If DCM is in SA or A-mode

3.2 Release of protective devices, signals 1 and 2

Channel 1 EX23:17+18

Channel 2 EX23:19+20

S,M,SA,A These signals are active when the safety gate of the DCM on the operator's side plus the fencing or safety gate on the rear side are closed (protection of the working area).

The sprayer may enter the die area exclusively when these signals are present. If the signals drop off while the sprayer is operating within the die area the unit must stop immediately and the drive units must be switched off (regulation from EN869). If the sprayer is placed on the left or right hand side of the DCM, the unit must be fenced in. The drive units have to be interlocked with the door of the fence.

3.3 Die open

SPFLSO1 EX23:27

S,M "Die open" signal

SA,A As S and M - additionally the shot has to be completed.

3.4 Release for entering into die area

SPFFOB EX23:28

S,M ----

SA,A "Die open"
 &
 "Robot out of collision area sprayer"
 &
 extraction completed
 &
 shot has to be completed
 &
 ejector was in forward position
 &
 corresponding safety gate open (when spraying not from the top)

In SA and A this signal will drop off as soon as the die spray unit signals "spray complete SPEND". It will be present again not before the next cycle of the DCM.

3.5 Ejector in rear position

SPAUSZ EX23:29

S,M,SA,A Is the limit switch signal

3.6 Command manual spraying

SPBZH EX23:30

S,SA,A ----

M Die open
 &
 robot out of collision area DCM
 &
 continuous actuation of the push-button "manual spraying"
 &
 Release from safety installation
 &
 "Spraying from the top or the equal safety gate is open"

On activation of push button „Manual Spray“, the safety gates will be moved to the positions as programmed in Spray position (OP3) before the spray cycle is started.

The signal will be reset in case any of the above conditions are no longer true or the sprayer signals base position SPIGS and spray complete SPEND was activated during the current spray cycle as well.

3.7 Command go to basic position

SPGSA EX23:31

S,SA,A ----

M "Die open"
 &
 Push button "Move plant to initial position" continuously activated.
 &
 Safety gates are in position for spraying (OP3)

Upon activation of this signal, the sprayer should move to base position using the shortest possible path.

The signal will be reset in case any of the above conditions are no longer true or the sprayer signals basic position SPIGS.

3.8 Die closed

SPFLSG EX23:32 (not in Version 21)

S,M,SA,A Corresponds with the limit-switch signal "die closed"

3.9 Operation mode setting

SPBAE **EX23:32 (Version 21)**

S If DCM is in S-mode

M,SA,A ----

3.10 Operation mode manual

SPBAH **EX23:33 (Version 21)**

M If DCM is in M-mode

S,SA,A ----

4 Signals: Die spray unit --> DCM Control

4.1 Die spray unit is in operation mode A / SA

SPIATA EX23:3

(Bridge 24V if unit cannot emit this signal)

Utilization: Not later than 2 sec. after switching over to SA or A this signal has to be present. Otherwise the operation mode changes to M and the corresponding message will be displayed on the screen. That makes sure that the unit is ready to start the cycle.

Message: "Die spray unit: not in operation mode A / SA" Q

4.2 Die spray unit in basic position

SPIGS EX23:4

Utilization: - Condition to change from S and M to SA or A (*)
- Part of sequence condition for "close die" (*)
- Lamp of push-button "go to basic position"
- Definition "end of cycle"
- Monitoring of collision area extractor - die spray unit

(*) Message: "Die spray unit: no basic position" NQ

Monitoring: - In operation mode SA and A this signal must change its status at least once during one DCM cycle. Otherwise the operation mode changes to M (at the end of cycle) and the corresponding message will be displayed on the screen.

Message: "Die spray unit: signal "in basic position" not cyclic actuated" Q

4.3 Spraying completed (Impulse: min. 200ms)

SPEND EX23:8

Utilization: - In operation mode A part of command "close die"
- Definition "end of cycle"
- Release for a manual operation after spraying (if selected)

4.4 Die spray unit fault priority 2

SPST2 EX23:10

Utilization: - Immediate change to SA and message display on screen
- Change from SA to M at the end of the cycle

Message: "Die spray unit: fault priority 2" NQ

(For example: spray medium level too low)

4.5 Emergency-off

Emergency-off die spray unit --> DCM, channel 1	EX23:12+14
Emergency-off die spray unit --> DCM, channel 2	EX23:21+22
Emergency-off message	EX23:13
Emergency-off DCM --> die spray unit, channel 1	EX23:15+16
Emergency-off DCM --> die spray unit, channel 2	EX23:23+24

The die spray unit is integrated in the 2-channel emergency-off circuit of the DCM control system via the interface. Since these two channels are monitored the peripheral units must also be equipped with a 2-channel Emergency-off.

The emergency-off buttons of the various peripheral devices are connected in a chain through which a redundant contactor combination is activated. The contactor combination drops off as soon as this chain is interrupted at any point. It switches off the control voltages of the various peripheral devices.

The DCM control system offers the possibility to display the corresponding emergency-off in plain text via an additional PLC input. If the sprayer is in emergency-off mode, the emergency-off voltage must be present at that input.

(see also signal allocation diagram of DCM)

4.6 Die spray unit out of die area

SPAFOB EX23:5 (Option)

Utilization:

- Condition to change from S and M to SA or A (*)
- Part of sequence condition for "close die" (*)
- Part of signal "release for extraction" for robot
- Lamp of push-button "go to basic position"
- Definition "end of cycle"
- Monitoring of collision area extractor - die spray unit

(*) Message: "Die spray unit: no basic position" Q

Monitoring:

- In operation mode SA and A this signal must change its status at least once during one DCM cycle.
- Otherwise the operation mode changes to M (at the end of cycle) and the corresponding message will be displayed on the screen.

Message: "Die spray unit: signal "out of die area" not cyclic actuated" Q

Buhler Interface for Extraction Robot (Ecoline)

Control:

Ecoline

Release

Identification:	Pages:	Author:	Dept.:	Reviewer:	Dept:	Date:
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Signal added : ENSEPAT, ENSE						
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First release						

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Content

1	Introduction.....	4
1.1	Purpose	4
1.2	Abbreviations	4
1.3	Electrical Specification	4
1.4	Signal exchange.....	4
1.4.1	DCM Control → Robot	4
1.4.2	Robot ---> DCM Control	4
1.5	Safety	5
1.5.1	Signal monitoring implemented in DCM control	5
1.5.2	Personal safety.....	5
1.5.3	Responsibility	5
2	Connection-description.....	6
2.1	Cable	6
2.2	Plug	7
3	Signals: DCM Control --> Robot.....	9
3.1	Operation mode automatic.....	9
3.2	Shot completed	9
3.3	Release for extraction	9
3.4	Cores out.....	9
3.5	Ejector in forward position.....	9
3.6	Command go to basic position.....	10
3.7	Fencing closed, signals 1 and 2.....	10
3.8	Release: Safety gate right or left closed, channel 1 and 2	10
3.9	Separation: out-of-tolerance, reject.....	10
3.10	Release for entering to insert.....	11
3.11	Release to insert	11
3.12	Release to leave insert position	11
3.13	Insertion selected	11
4	Signals: Robot --> DCM Control.....	12
4.1	Robot is in operation mode A / SA.....	12
4.2	Robot out of collision area DCM	12
4.3	Command ejector forward.....	12
4.4	Robot has extracted (Impulse: min. 200ms, max. 2 sec.).....	12
4.5	Robot fault priority 2	13
4.6	Emergency-off.....	13
4.7	Insert in position	13
4.8	Robot out of collision area cores.....	13

1 Introduction

1.1 Purpose

The interface is suitable for:

- Extraction with robot
- Extraction from the left*, the right* or the top*
- Separation with robot
- Insertion with robot

The interface is not suitable for:

- Spraying with robot (-> Special interface)
- Communication with special peripheral units (-> Special interface)

*) Has to be specified with your order

1.2 Abbreviations

S	Operation mode setting
M	Operation mode manual
SA	Operation mode semi-automatic
A	Operation mode automatic
DCM	Die casting machine
Q	Fault message can be acknowledged on the keyboard of the DCM control
NQ	Fault message cannot be acknowledged

1.3 Electrical Specification

Output signals of DCM control:

Potential free relays contactors max. 2A / max. AC 250V / DC 30V min. 10mA/5V.

Input signals of DCM control:

min. 10mA / 24V DC (0V till +5V are considered 0V and +15V till +30V are considered 24V.

Control voltage from DCM).

In the DCM control cabinet all signals are wired on plugs (see also signal allocation diagram of DCM).

1.4 Signal exchange

1.4.1 DCM Control → Robot

All output signals are only present, when "Extraction" is selected on screen.

1.4.2 Robot ---> DCM Control

Except the signal "Robot out of collision area DCM" all input signals are only processed by the DCM control, when "Extraction" is selected on screen.

1.5 Safety

1.5.1 Signal monitoring implemented in DCM control

Some signals are monitored for cyclic actuation, e.g. these signals must change their status at least once during one DCM cycle. Otherwise the installation changes to manual mode (end installation cycle) and a corresponding message will be displayed on the screen (see signal description).

1.5.2 Personal safety

According to safety regulations the operator must not be endangered by movements of the unit. (--> place a fence). Apertures in the protecting system of the working place, such as doors in the fencing and the safety gate on the operator's side of the DCM, must be linked with the robot. To this end potential-free signals (2-channel) have been provided on the interface.

Signal "Fencing closed" (for details see 3.7.):

The two potential-free relay contacts must be linked in such a manner that the drive units of the robot are switched off when the fencing is open.

Signal "Safety gate right or left closed" (for details see 3.8.):

When the result of the safety analyses demands the extraction unit is only working when these signals are on, the unit must stop immediately and the drive units must be switched off, if these signals drop out.

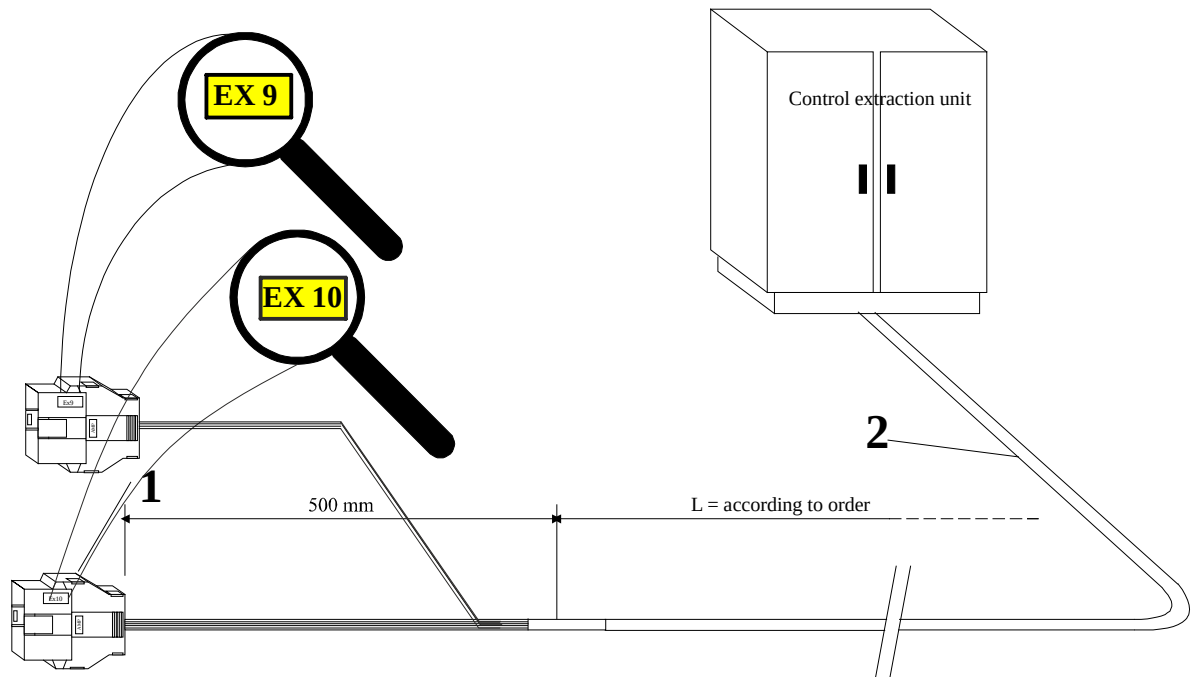
1.5.3 Responsibility

If the unit is furnished by the customer, he carries the responsibility for conformation of the unit to regulations specific the country (e.g. CE) and that it is linked to the installation in accordance (see also 1.5.2. personal safety).

Before setting into operation the effective location of the unit (to the left or right) corresponds with the position as predetermined in the DCM (symbol of robot on page "Information" = initial image). If this is not the case this must be altered immediately (contact customer service).

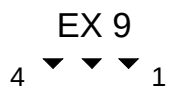
2 Connection-description

2.1 Cable



The interface cable (1+2) will be manufactured by the producer of the periphery gear. A plug package (1) can be ordered from Bühler AG.

2.2 Plug



- 1: 24 Volt from DCM
- 2: GND from DCM
- 3: ENIATA: Robot is in operation mode A / SA
- 4: ENAKD1: Robot out of collision area DCM
- 5: ENBAUV: Command ejector forward
- 6: ENHE: Robot has extracted
- 7: ENST2: Robot in fault priority 2
- 8:
- 9: Robot supply voltage for dry contacts
- 10: ENBAA: Operation mode automatic
- 11: ENFE1: Release for extraction
- 12: ENASAKA: cores out
- 13: ENAUSV: Ejector in forward position
- 14: ENRENDG: shot completed
- 15: ENGSA: Command go to basic position
- 16: ENINPOS: Insert in position
- 17: Emergency-off robot → DCM, channel 1
- 18: Emergency-off message → DCM
- 19: Emergency-off robot → DCM, channel 1
- 20: Emergency-off DCM → robot, channel 1
- 21: Emergency-off DCM → robot, channel 1
- 22: Emergency-off DCM → robot, channel 2
- 23: Emergency-off DCM → robot, channel 2
- 24: Fencing closed, signal 1
- 25: Fencing closed, signal 1
- 26: Fencing closed, signal 2
- 27: Fencing closed, signal 2
- 28: Release: Safety gate right or left closed, channel 1
- 29: Release: Safety gate right or left closed, channel 1
- 30: Release: Safety gate right or left closed, channel 2
- 31: Release: Safety gate right or left closed, channel 2
- 32: ENAKK: Robot out of collision area cores
- 33: Emergency-off robot → DCM, channel 2
- 34: Emergency-off robot → DCM, channel 2
- 35: ENSEPAT: Separation out of tolerance
- 36: ENSEPA: Separating reject

EX 10
4 ▼ ▼ ▼ 1

36 33

- 1: Robot supply voltage for dry contacts
- 2:
- 3:
- 4:
- 5:
- 6:
- 7:
- 8:
- 9:
- 10:
- 11:
- 12:
- 13:
- 14:
- 15:
- 16:
- 17:
- 18:
- 19:
- 20:
- 21:
- 22:
- 23: ENFIN1: Release for entering to insert signal 1
- 24:
- 25: ENFINS: Release to insert
- 26: ENFOUT: Release to leave insert position
- 27:
- 28:
- 29:
- 30: ENINSON: Insertion selected on DCM
- 31:
- 32:
- 33:
- 34:
- 35:
- 36:

3 Signals: DCM Control --> Robot

3.1 Operation mode automatic

ENBAA EX9:10

S,M ----

SA,A If DCM is in SA or A-mode

3.2 Shot completed

ENRENDG EX9:14

S ----

M,SA,A Is present after a shot is completed until the die is open.

3.3 Release for extraction

ENFE1 EX9:11

S,M "Die open" or if S:Rob sim "die and extraction simultaneous" =1:
programmable stroke point s LsimRob

&

corresponding safety gate open

&

release of die spray unit

SA,A As S and M - additionally the shot has to be completed.

3.4 Cores out

ENASAKA EX9:12

S,M,SA,A Is a collection of the limit switch signals "core out" of all selected cores

This signal has to be analysed in addition if the casting cannot be approached before
the cores are moved out

3.5 Ejector in forward position

ENASV EX9:13

S,M,SA,A Corresponds with the limit switch signal of "ejector forward"

3.6 Command go to basic position

ENGSA EX9:15

S,SA,A ----

M The signal is present :
 - when the safety gates are in a suitable position for extraction
 - as long as the robot reports being inside the die area (not ENAKD1)
 - as the push-button "Move plant to initial position" at the DCM operator control panel is pressed
 - as long as the sprayer is out of the die area

Inside the die area the robot is only allowed to move as long as this signal is present.

3.7 Fencing closed, signals 1 and 2

Fencing closed, signal 1 EX9:24+25

Fencing closed, signal 2 EX9:26+27

S,M,SA,A These signals are active as long as the fence is closed

The two potential-free relay contacts must be linked in such a manner that the drive units of the robot are switched off when the fencing is open.

3.8 Release: Safety gate right or left closed, channel 1 and 2

Release: Safety gate right or left closed, channel 1 EX9:28+29

Release: Safety gate right or left closed, channel 2 EX9:30+31

S,M,SA,A These signals are active as long as the right or the left safety gate (minimum one) of the DCM is closed. (Protection of working area of robot.)

When the result of the safety analyses demands the extraction unit is only working when these signals are on, the unit must stop immediately and the drive units must be switched off, if these signals drop out.

3.9 Separation: out-of-tolerance, reject

ENSEPAT EX9:35 (optional)

ENSEPA EX9:36 (optional)

Further information on the subject separation and quality characteristics can be found in the operation manual of the die casting machine

S ----

M,SA,A This signal is only present from the message "robot has extracted" until the signal "shot completed" in the following cycle

Reason for this late time of output:

With the signal "robot has extracted" the robot reports that the shot-completeness control is O.K. If it is not O.K. the robot indicates "fault priority 2" and the installation changes to manual mode. In this way the quality characteristic "good" will change to "reject".

3.10 Release for entering to insert

(Only available with option ,Insertion with robot')

ENFIN1 EX10:23

M,SA,A These signals are active if ,Insertion with robot' is selected on DCM, die is open, the appropriate safety gate is open and the sprayer signals out of die area. In addition the start-up program can not to be active and the screen switch S: PS nins (Production start without insertion) is used to enable/disable insertion during production start.

The signal 'Release for entering to insert' allows the robot to advance to a wait position ,Clear of cores'.

3.11 Release to insert

(Only available with option ,Insertion with robot')

ENFINS EX10:25

M,SA,A This signal is active after the robot signals ,Wait position clear off cores'. It allows the robot to insert the part.

3.12 Release to leave insert position

(Only available with option ,Insertion with robot')

ENFOUT EX10:26

M,SA,A The release to leave the insert position is available as soon as the robot signals part inserted (ENINPOS) and if core(s) are used for clamping, the CNP has moved on to the INS step EXTRACT.

3.13 Insertion selected

ENINSON EX10:30

S,M,SA,A This signal is active as long as insertion with robot is selected on the HMI of the DCM.

4 Signals: Robot --> DCM Control

4.1 Robot is in operation mode A / SA

ENIATA EX9:3

(Bridge 24V if unit can not emit this signal)

Utilization: Not later than 2 sec. after switching over to SA or A this signal has to be present. Otherwise the installation changes to M and the corresponding message will be displayed on the screen.
That makes sure that the unit is ready to start the cycle.

Message: "Extractor: not in operation mode A / SA" Q

4.2 Robot out of collision area DCM

ENAKD1 EX9:4

Utilization:

- Condition to change from S and M to SA or A
- Part of sequence condition for "close die"
- Part of command and sequence condition for "close corresponding safety gate"
- Part of signal "release for entering into die area" for die spray unit
- Lamp of push-button "go to basic position"
- Definition "end of cycle"
- Monitoring of collision area extractor - die spray unit

4.3 Command ejector forward

ENBAUV EX9:5

Utilization: Part of command and sequence condition "ejector forward movement"
but only when the robot is in the collision area of the DCM.

Message: "Extractor: no release for ejector" NQ will be displayed if this signal is not set on time.

4.4 Robot has extracted (Impulse: min. 200ms, max. 2 sec.)

ENHE EX9:9

This signal has to be present if the robot has moved out and the shot-completeness control reports O.K. If it does not report O.K. the robot has to emit "fault priority 2" and indicate "fault shot-completeness control" on its own display. This signal must be reset before the next cycle is started, thus the max. limit of 2 sec.

Utilization:

- Final condition for „time out extraction process not ended“
- In A mode condition for cycle start, except during 1. cycle
- Condition to set separation signals and incrementing cycle counter.
- Part of signal "release for entering into die area" for die spray unit

4.5 Robot fault priority 2

ENST2 EX9:13

Utilization: - Immediate change to SA and message display on screen
- Change from SA to M at the end of installation cycle

Message: "Extractor: fault priority 2" NQ

4.6 Emergency-off

Emergency-off robot--> DCM, channel 1	EX9:17+19
Emergency-off robot--> DCM, channel 2	EX9:33+34
Emergency-off message --> DCM	EX9:18
Emergency-off DCM--> robot, channel 1	EX9:20+21
Emergency-off DCM--> robot, channel 2	EX9:22+23

The robot is integrated in the 2-channel emergency-off circuit of the DCM control system via the robot interface. Since these two channels are monitored the peripheral units must also be equipped with a 2-channel Emergency-off.

The emergency-off buttons of the various peripheral devices are connected in a chain through which a redundant contactor combination is activated. The contactor combination drops off as soon as this chain is interrupted at any point. It switches off the control voltages of the various peripheral devices.

The DCM control system offers the possibility to display the corresponding emergency-off in plain text via an additional PLC input. If the robot is in emergency-off mode, the emergency-off voltage must be present at that input (see also signal allocation diagram of DCM).

4.7 Insert in position

(Only available with option „Insertion with robot“)

ENINPOS EX9:16

This signal is used to set the condition „Insert in position“ for the DCM.

Utilization: Setting of the condition 'Robot at insert position' (ENPOSEL), i.e. cores could be moved at this point to secure the part.

4.8 Robot out of collision area cores

(Only available with option „Insertion with robot“)

ENAKK EX9:32

The robot is outside of the collision area with the cores.

Utilization: - Release (in addition to ENAKD) for 'Safety gate closing'
- Terminates the action 'Move in/out for insertion' resp. move out after insertion'.

Buhler Interface for Ladling unit (Ecoline)

Control:

Ecoline

Release

Identification:	Pages:	Author:	Dept.:	Reviewer:	Dept:	Date:
GEB19000en.ECO.1E	11	E. Guyon	DF6	M. Louis	DF6	08.10.2014
		Enhancement with fencing metal feeding channels (Chapter 3.5)				
GEB19000en.ECO.1D	11	M. Louis	DF7	none		05.11.2013
		Correction of address, logo, electrical specifications, picture 2.1, text 1.2, 3.6, 3.7, 3.8, 3.9, 4.2				
GEB19000en.1C	11	E. Guyon	DF6	M. Louis	DF7	08.08.2012
		Die and release for dosing simultaneous				
GEB19000en.1B	11	E. Guyon	DF6	M. Louis	DF7	15.04.2010
		Signal added: MZSDM, MZREGS, MZFLSG, MZFLSO				
GEB19000en.1A	11	M. Lütolf	DF6	M. Louis	DF7	09.09.2009
		First release				

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Content

1	Introduction.....	4
1.1	Purpose	4
1.2	Abbreviations	4
1.3	Electrical Specification	4
1.4	Signal exchange.....	4
1.4.1	DCM Control → Ladling Unit	4
1.4.2	Ladling Unit → DCM Control	4
1.5	Safety	5
1.5.1	Signal monitoring implemented in DCM control	5
1.5.2	Responsibility	5
2	Connection-description.....	6
2.1	Cable	6
2.2	Plug	7
3	Signals: DCM Control --> Ladling Unit	8
3.1	Start manual dosing	8
3.2	Operation mode automatic.....	8
3.3	Start ladling unit	8
3.4	Die closing active	8
3.5	Fencing metal feeding (new from 2014.10.08)	8
3.6	Release fill in.....	9
3.7	Command go to basic position.....	9
3.8	Die open	10
3.9	Plunger in rear position	10
3.10	Die closed / Die not closed (option)	10
4	Signals: Ladling Unit --> DCM Control	11
4.1	Ladling unit is in operation mode A / SA	11
4.2	Command shot forward (Impulse: min. 200ms)	11
4.3	Ladling unit fault priority 2	11
4.4	Emergency-off.....	11

1 Introduction

1.1 Purpose

The interface is suitable for:

- Metal feeding

1.2 Abbreviations

S	Operation mode setting
M	Operation mode manual
SA	Operation mode semi-automatic
A	Operation mode automatic
DCM	Die casting machine
Q	Fault message can be acknowledged on the keyboard of the DCM control
NQ	Fault message cannot be acknowledged
Version 21	Indication of the second release of the interface with additional signals

1.3 Electrical Specification

Output signals of DCM control:

Potential free relays contactors max. 2A / max. AC 250V / DC 30V min. 10mA/5V.

Input signals of DCM control:

min. 10mA / 24V DC (0V till +5V are considered 0V and +15V till +30V are considered 24V. Control voltage from DCM).

In the DCM control cabinet all signals are wired on plugs (see also signal allocation diagram of DCM).

1.4 Signal exchange

1.4.1 DCM Control → Ladling Unit

All output signals are only present, when "Ladling unit" is selected on screen.

1.4.2 Ladling Unit → DCM Control

All input signals are only processed by the DCM control, when "Ladling unit" is selected on screen.

1.5 Safety

1.5.1 Signal monitoring implemented in DCM control

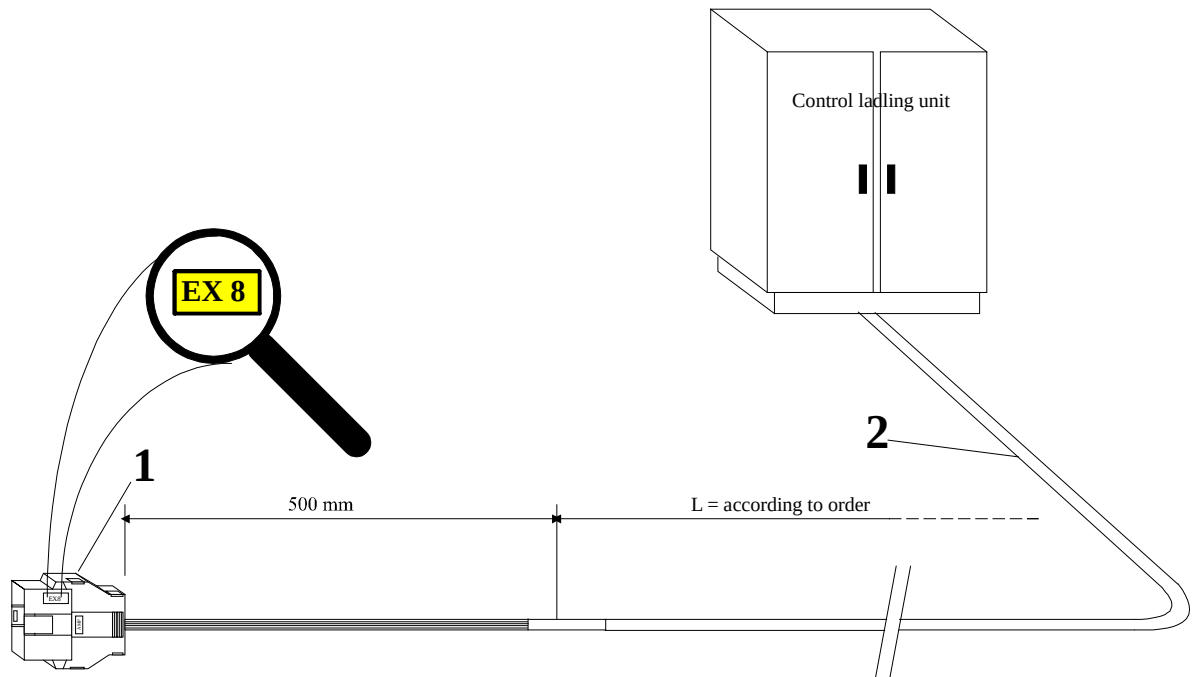
Some signals are monitored for cyclic actuation, e.g. these signals must change their status at least once during one DCM cycle. Otherwise the installation changes to manual mode (end installation cycle) and a corresponding message will be displayed on the screen (see signal description).

1.5.2 Responsibility

If the unit is furnished by the customer, he carries the responsibility for conformation of the unit to regulations specific the country (e.g. CE) and that it is linked to the installation in accordance.

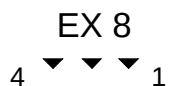
2 Connection-description

2.1 Cable



The interface cable (1+2) will be manufactured by the producer of the periphery gear. A plug package (1) can be ordered from "Bühler AG".

2.2 Plug



- 1: 24 Volt from DCM
- 2: GND from DCM
- 3: MZIATA: Ladling unit is in operation mode A / SA
- 4:
- 5: MZFRE1: Command shot forward
- 6:
- 7:
- 8: MZST2: Ladling unit fault priority 2
- 9:
- 10: MZFLSO: Die open
- 11: Emergency-off ladling unit → DCM, channel 1
- 12: Emergency-off message → DCM
- 13: Emergency-off ladling unit → DCM, channel 1
- 14: Emergency-off DCM → ladling unit, channel 1
- 15: Emergency-off DCM → ladling unit, channel 1
- 16: 24 Volt from dosing unit
- 17: MZSDM: Start manual dosing
- 18:
- 19: MZBAA: Operating mode automatic
- 20: MZST: Start ladling unit
- 21: MZFLZU: Die closing active
- 22:
- 23:
- 24: MZFE1: Release fill in
- 25: MZREGS: Plunger in rear position
- 26: Fencing metal feeding, channel 1
- 27: Fencing metal feeding, channel 1
- 28: MZFLSG: Die closed / MZNFLSG: Die not closed (option)
- 29: MZGSA: Command go to basic position
- 30: Fencing metal feeding, channel 2
- 31: Fencing metal feeding, channel 2
- 32:
- 33: Emergency-off ladling unit → DCM, channel 2
- 34: Emergency-off ladling unit → DCM, channel 2
- 35: Emergency-off DCM → ladling unit, channel 2
- 36: Emergency-off DCM → ladling unit, channel 2

3 **Signals: DCM Control --> Ladling Unit**

3.1 **Start manual dosing**

MZSDM EX8:17 (Version 21)

S,SA,A ----

M This signal is present as long as the push-button "start manual dosing" at the DCM operator control panel is pressed.

3.2 **Operation mode automatic**

MZBAA EX8:19

S,M ----

SA,A If DCM is in SA- or A-mode

3.3 **Start ladling unit**

MZST EX8:20

S,M ----

SA,A This signal is present after each cycle start until "plunger in rear position" drops off.

3.4 **Die closing active**

MZFLZU EX8:21

S ----

M,SA,A This signal is present during the entire closing movement of the die and as long as the limit switch "Die closed" is actuated.

3.5 **Fencing metal feeding (new from 2014.10.08)**

(Only available with option 'fencing metal feeding')

Fencing metal feeding, channel 1 EX8:26+27

Fencing metal feeding, channel 2 EX8:30+31

S,M,SA,A These signals are active as long as the fence is closed

The two potential-free relay contacts must be linked in such a manner that the drive units of the ladling unit are switched off when the fencing is open.

3.6 Release fill in

MZFE1	EX8:24
S	----
M	Button 61-ES88 activated & plunger in rear position & die closed & Lockmat OK & cores moved in & shot ready (accumulator loaded) & release of safety equipment DCM & not "shot completed"
SA,A	S:LsimDos on HMI = 0 or not available & plunger in rear position & die closed & Lockmat OK & cores moved in & shot ready (accumulator loaded) & release of safety equipment DCM & not "shot completed"
OR	
SA,A	S:LsimDos on HMI = 1 plunger in rear position & die position shorter than s LsimDos & release of safety equipment DCM & not "shot completed"

3.7 Command go to basic position

MZGSA	EX8:29
S,SA,A	----
M	This signal is present as long as the push-button "Move plant to initial position" at the DCM operator control panel is pressed. The unit must move only as long as the switch is pressed.

3.8 **Die open**

MZFLSO EX8:10 (Version 21)

S,M,SA,A Corresponds with the limit switch signal of "die open"

3.9 **Plunger in rear position**

MZREGS EX8:25 (Version 21)

S,M,SA,A Corresponds with the limit switch signal of "plunger in rear position"

3.10 **Die closed / Die not closed (option)**

A) Die closed

MZFLSG EX8:28 (Version 21)

S,M,SA,A This signal is on if the limit switch signal is on and Lockmat is OK

B) Die not closed (option)

MZNFLSG EX8:28 (Version 21)

With this option the signal "die closed" is replaced by "die not closed". There are units (e.g. Rimrock) which need this signal instead of "die closed".

S,M,SA,A Corresponds with the inverse limit switch signal of "die closed"

4 Signals: Ladling Unit --> DCM Control

4.1 Ladling unit is in operation mode A / SA

MZIATA EX8:3

(Bridge 24V if unit cannot emit this signal)

Utilization: Not later than 2 sec. after switching over to SA or A this signal has to be present. Otherwise the installation changes to M and the corresponding message will be displayed on the screen.
That makes sure that the unit is ready to start the cycle.

Message: "Ladling unit: not in operation mode A / SA" Q

4.2 Command shot forward (Impulse: min. 200ms)

MZFRE1 EX8:5

Utilization: - To change from S and M to SA or A, this signal mustn't be present
- In operation mode SA and A part of the command "Shot forward"

4.3 Ladling unit fault priority 2

MZST2 EX8:8

Utilization: - Immediate change to SA and message display on screen
- Change from SA to M at the end of the installation cycle

Message: "Ladling unit: fault priority 2" NQ

4.4 Emergency-off

Emergency-off ladling unit → DCM, channel 1	EX8:11+13
Emergency-off ladling unit → DCM, channel 2	EX8:33+34
Emergency-off message → DCM	EX8:12
Emergency-off DCM → ladling unit, channel 1	EX8:14+15
Emergency-off DCM → ladling unit, channel 2	EX8:35+36

The ladling unit is integrated in the 2 channel emergency-off circuit of the DCM control system via the interface. Since these two channels are monitored the peripheral units must also be equipped with a 2-channel Emergency-OFF.

The emergency-off buttons of the various peripheral devices are connected in a chain through which a redundant contactor combination is activated. The contactor combination drops off as soon as this chain is interrupted at any point. It switches off the control voltages of the various peripheral devices.

The DCM control system offers the possibility to display the corresponding emergency-off in plain text via an additional PLC input. If the ladling unit is in emergency-off mode, the emergency-off voltage must be present at that input.

(See also signal allocation diagram of DCM)

Buhler Interface for Vacuum (Ecoline)

Control:

Ecoline

Release

Identification:	Pages:	Author:	Dept.:	Reviewer:	Dept:	Date:
GEB19600en.1D Correction of output voltage, suitable devices, Bühler logo, picture 2.1	8	M. Louis	DF7	unreviewed		31.10.2013
GEB19600en.1C Enhancement with signals : VAGUT, VASTAT, VAFLSO, VAFLSG	8	E. Guyon	DF6	M. Louis	DF7	26.06.2013
GEB19600en.1B Enhancement with Emergency stop signal	8	E. Guyon	DF6	M. Louis	DF7	04.02.2012
GEB19600en.1A First release	8	M. Lütolf	DF6	M. Louis	DF7	09.09.2009

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Content

1	Introduction.....	4
1.1	Purpose	4
1.2	Abbreviations	4
1.3	Electrical Specification	4
1.4	Signal exchange.....	4
1.4.1	DCM Control → Vacuum.....	4
1.4.2	Vacuum → DCM-Control.....	4
1.5	Safety	4
1.5.1	Responsibilities	4
2	Connection-description.....	5
2.1	Cable	5
2.2	Plug	6
3	Signals: DCM Control --> Vacuum	7
3.1	Auxiliary supply	7
3.2	With / without vacuum	7
3.3	Activation vacuum valve.....	7
3.4	Vacuum stop	7
3.5	Activation blow valve.....	8
3.6	Die open	8
3.7	Die closed.....	8
4	Signals: Vacuum --> DCM Control	9
4.1	Power supply.....	9
4.2	Fault: fault priority 1.....	9
4.3	Fault: fault priority 2 (End of cycle)	9
4.4	Vacuum ready	9
4.5	E-Stop / E-Stop indication	9
4.6	Fault: Out of tolerance (Reject GEVA14).....	10
4.7	Vacuum OK (GEVA14)	10

1 Introduction

1.1 Purpose

The interface is suitable for:

- simple Vacuum (i.e. Provac, Fondarex)

1.2 Abbreviations

S	Operation mode setting
M	Operation mode manual
SA	Operation mode semi-automatic
A	Operation mode automatic
DCM	Die casting machine
Q	Fault message can be acknowledged on the keyboard of the DCM control
NQ	Fault message cannot be acknowledged

1.3 Electrical Specification

Output signals of DCM control:

Potential free relays contactors max. 2A / max. AC 250V / DC 30V min. 10mA/5V.

Input signals of DCM control:

min. 10mA / 24V DC (0V till +5V are considered 0V and +15V till +30V are considered 24V.

Control voltage from DCM).

1.4 Signal exchange

1.4.1 DCM Control → Vacuum

Output signals are only present, when "vacuum unit" = 1 is selected on screen. Exceptions are mentioned in the specification of the respective signal.

1.4.2 Vacuum → DCM-Control

All input signals are only processed by the DCM control, when "Vacuum unit" = 1 is selected on screen.

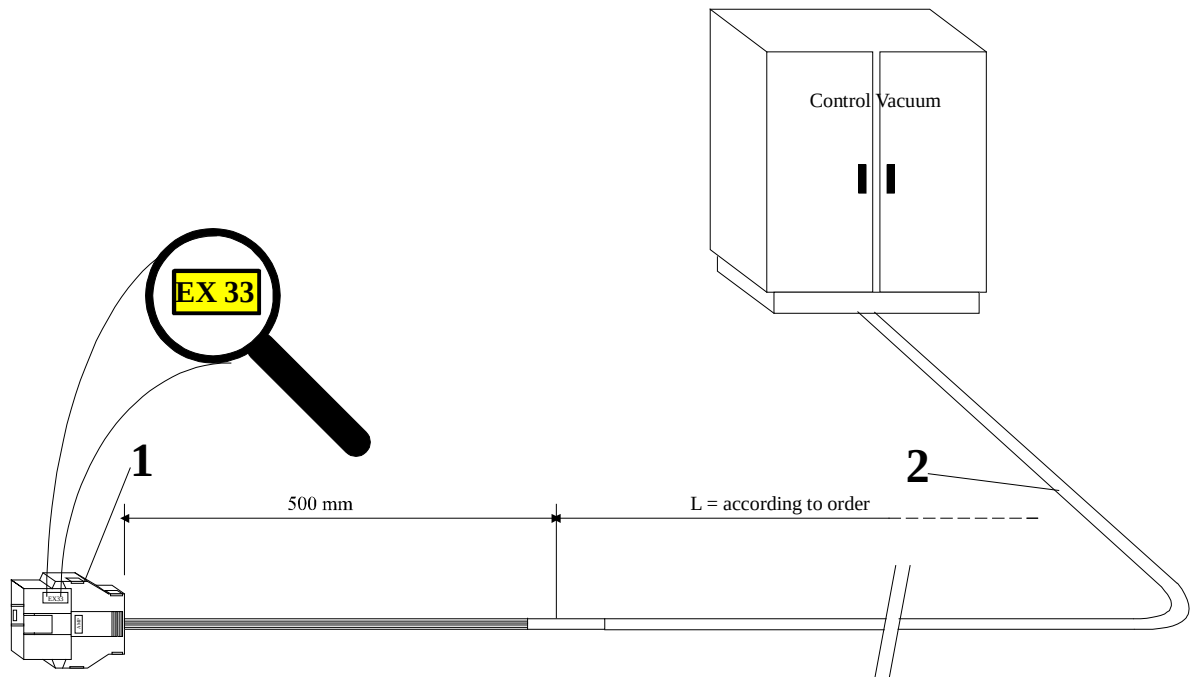
1.5 Safety

1.5.1 Responsibilities

In case the peripheral equipment is supplied by a third party, they will be responsible, that the equipment as well as the installation is executed according to local code.

2 Connection-description

2.1 Cable



The interface cable (1+2) will be manufactured by the producer of the periphery gear. A plug package (1) can be ordered from "Bühler AG".

2.2 Plug

EX 33
4 ▼ ▼ ▼ 1

36 33

- 1: Power supply 24 Volt from DCM
- 2: Power supply GND from DCM
- 3: E Stop Vacuum --> DCM, Channel 1
- 4: E-Stop Indication
- 5: E Stop Vacuum --> DCM, Channel 1
- 6: E Stop Vacuum --> DCM, Channel 2
- 7: E Stop Vacuum --> DCM, Channel 2
- 8: E-Stop DCM --> Vacuum, Channel 1 (HW V2)
- 9: E-Stop DCM --> Vacuum, Channel 1 (HW V2)
- 10: E-Stop DCM --> Vacuum, Channel 2 (HW V2)
- 11: E-Stop DCM --> Vacuum, Channel 2 (HW V2)
- 13: VAST1: fault, priority 1
- 14: VAST2: fault, priority 2 (End of cycle)
- 15: VASTAT: fault, Out of tolerance
- 16: Auxiliary supply
- 17: VAST: Activation vacuum valve
- 18: VABL: Activation blow valve
- 19: VASP: Vacuum stop
- 20: VAMIT: With / without vacuum
- 21: VAFLSO: Die open
- 22: VAGUT: Vacuum OK
- 23: VAIBB: Vacuum ready
- 24: VAFLSG: Die closed

3 Signals: DCM Control --> Vacuum

3.1 Auxiliary supply

EX33:16 Ext. Supply Voltage from Vacuum-Control

3.2 With / without vacuum

VAMIT **EX33:20**

S Continuous signal, as long as „vacuum“ = 1 is selected on the HMI

M Continuous signal, as long as:
 „vacuum“ = 1 is selected on the HMI
 &
 "Vacuuming handmode" = 1 is selected on the HMI

SA,A Continuous signal, as long as:
 „vacuum“ = 1 is selected on the HMI

3.3 Activation vacuum valve

VAST **EX33:17**

S ----

M,SA,A This signal is activated as soon as the signal "**VAMIT**" is active and the plunger has, during a shot (not plunger forward), reached the programmed stroke point sVacBeg until

- the Timer t Vac has expired * or
- the programmable stroke point sVacEnd has been reached *

*) End condition is selectable on the HMI.

Attention:

M,SA,A If "Selection: vacuum type" = Fondarex VACUPAC then this signal is also generated, as described above, when the vacuum interface is deselected, "Vacuum" = 0. "VAMIT" does not have to be active.

3.4 Vacuum stop

VASP **EX33:19**

M,SA,A After output „**VAST**“ is deactivated, this signal will be active for 4 seconds.

3.5 **Activation blow valve**

VABL EX33:18
S ----
M,SA,A This signal is active as long as “**VAMIT**” is active and the programmable Timer t Blow has not expired, but will be cancelled latest at cycle start.
 – t Blow is activated, after the programmable delay t Blow R has expired
 – t Blow R is activated after a shot has been made and the die is „open“

3.6 **Die open**

VAFLSO EX33:21
S,M,SA,A This signal is active if “Vacuum” = 1 is selected and as soon as the die is „open“.

3.7 **Die closed**

VAFLSG EX33:24
S -----
M,SA,A This signal is active if “VAMIT” is active and as soon as the die is „closed“, until vacuum stop “VASP”.

4 Signals: Vacuum --> DCM Control

4.1 Power supply

EX33:1 DCM 24 Volt supply

EX33:2 DCM 0 Volt/ground

4.2 Fault: fault priority 1

VAST1 **EX33:13**

Usage: Immediate switch over from A / SA to M

Fault text: "Vacuum: Fault priority 1" NQ

4.3 Fault: fault priority 2 (End of cycle)

VAST2 **EX33:14**

If the vacuum system detects a fault at any time during the cycle, the machine will finish the cycle and then stop.

Usage: - Immediate switch over from A to SA and fault indication on HMI
- Switch over from SA to M at end of machine/cell cycle

Fault text: "Vacuum: Fault priority 2" NQ

4.4 Vacuum ready

VAIBB **EX33:23**

(Bridge 24V if unit cannot emit this signal)

The signal „Vacuum ready“ (VAIBB) indicates to the DCM, that the vacuum system is operational. This signal is only processed if „With/without vacuum“ (VAMIT) is active and the DCM OP mode is S/M/SA/A.

Usage: DCM switches 1 sec. after loss of signal VAIBB to SA and at end of cycle to M.

Fault text: "Vacuum: System not ready" Q

4.5 E-Stop / E-Stop indication

E Stop Vacuum --> DCM, Channel 1 **EX33:3+5**

E-Stop Vacuum --> DCM, Channel 2 **EX33:6+7**

E-Stop Indication **EX33:4**

E-Stop DCM --> Vacuum, Channel 1 **EX33:8+9 (HW V2)**

E-Stop DCM --> Vacuum, Channel 2 **EX33:10+11 (HW V2)**

By connecting the vacuum unit to the DCM vacuum interface, it becomes an integral part of the DCM 2 channel E-Stop circuit. Since both channels are actively monitored, the vacuum unit has to be equipped with a 2 channel E-Stop circuit as well.

The E-Stop push buttons of the DCM and the peripheral units are daisy chained together and control a fail safe contactor group. This group will deactivate as soon as the daisy chain is broken. The contactors will then interrupt the control voltages to any peripheral equipment integrated into the circuit.

The DCM-control will show a peripheral unit specific fault message on the HMI, if the appropriate PLC input is activated. In case a vacuum E-Stop is present, this input has to be activated (24VDC from DCM circuit).

For specifics on the wiring requirements, please consult the electrical schematics for the DCM.

4.6 Fault: Out of tolerance (Reject GEVA14)

VASTAT EX33:15

If the vacuum system detects a condition which would compromise the quality of the casting, it can relay that fault to the DCM by activating this input (Impulse: min. 200ms). However: This impulse has to be present while the die is still closed!

Usage: The monitoring system will flag the casting as „Out of tolerance“ (i.e. any documentation generated will reflect this quality attribute) and the extraction robot will place the casting in the programmed drop off position for out of tolerance castings.

Fault text: "Vacuum: Out of tolerance" NQ

Attention: If the option ‚Extension separation signals GEVA14‘ is active the following specification is valid:

Usage: The monitoring system will flag the casting as „reject“ and the extraction robot will place the casting accordingly.

Fault text: "Vacuum: rejects" NQ

4.7 Vacuum OK (GEVA14)

(Only available with option ‚Extension separation signals GEVA14‘)

VAGUT EX33:22

This signal indicates to the DCM that a vacuum cycle has been executed. It is only processed by the DCM if the option GEVA14 is active and the DCM OP mode is SA/A.

Usage: Parts produced without signal „Vacuum OK“ present will be separated as reject parts.

Fault text: "Vacuum: No vacuum" NQ

Network Dat@net

Contents

1	Basics	3
1.1	Machine Network	3
1.2	Transmission Protocol	3
1.3	Connection	3
2	System Requirements	4
2.1	Hardware	4
2.1.1	Server	4
2.1.2	PC (optional if remote control)	4
2.1.3	Printer (optional)	4
2.1.4	Modem / Router (optional if remote control)	4
2.1.5	Cabling	4
2.2	Software	5
2.2.1	Server	5
2.2.2	PC	5
3	Preparation	6
3.1	Responsibility of Customer	6
3.2	Responsibility of Bühler AG	6
4	Startup	7
4.1	Data Storage on the Server	7
4.1.1	Tasks of Customer	7
4.1.2	Tasks of Bühler AG	7
4.2	Remote control (Office)	7
4.2.1	Tasks of Customer	7
4.2.2	Tasks of Bühler AG	7

4.3	Remote Diagnostics	8
4.3.1	Tasks of Customer	8
4.3.2	Tasks of Bühler AG	8
5	Configuration of the Machine	9
5.1	HMI Settings	9
5.2	Customer's Settings	10
6	Software installation	11
6.1	Jave Runtime Environment	11
6.1.1	Installation	11
6.2.	Settings	13
6.2.1	Java Runtime Enviroment without Proxy-Server	13
6.2.2	Java Runtime Enviroment with Proxy-Server	14
6.2.3	Internet-Explorer without Proxy-Server.	16
6.2.4	Internet-Explorer without Proxy-Server.	17
7	Remote control	19
7.1	By Internet Browser (e.g. Windows Internet Explorer)	19
7.2	By JavaWebstart	22

1 Basics

1.1 Machine Network

The machine network is based on the principle of the client server. This means that all machines are clients of the network server. On the server, individual shares have to be installed for each connected machine in order to save predefined data through the HMI on these individual shares.

In order to guarantee the physical separation with the customer's network, the server must be equipped with two network cards.

A printer connected of the machine network can be selected from each machine and can be used for the printout of documents.

With the computer, which is linked to the machine network, one can access to the machines.

Bühler AG recommends to use this computer only for remote control, monitoring and for diagnostics of the machines.

With a software for remote control, which is establishing a secure connection through the internet, the customer service department of Bühler AG can take over the remote control via the host Computer.

1.2 Transmission Protocol

The TCP/IP protocol is used as transmission protocol.

1.3 Connection

In the standard scope of supply of the Datanet control, a connector socket TPI (Twisted Pair Interface) is mounted in the control unit where an 8-pins RJ 45 connector (MOD-JACK) can be connected.

2 System Requirements

2.1 Hardware

2.1.1 Server

- Processor, at least Pentium 3
- Working memory at least 512 MB
- Harddrive with at least 10 GByte
- Network card Ethernet 10/100 Mbit LAN NIC according IEEE 802.3 with RJ45-connections

2.1.2 PC (optional if remote control)

- Processor, at least Pentium 3
- Working memory at least 512 MB
- Network card Ethernet 10/100 Mbit LAN NIC according IEEE 802.3 with RJ45-connections
- If remote maintenance, CF flash card reader, internal or external

2.1.3 Printer (optional)

- Printer language PS (Post-Script).

2.1.4 Modem / Router (optional if remote control)

- ADSL Modem or Router

2.1.5 Cabling

- We recommend to use fiber optic cables in the foundry.
- The wiring length between the machine and the Hub/Server must not be longer than 40m.

2.2 Software

2.2.1 Server

- Server-Software (e.g. Windows 2003 Server)

2.2.2 PC

- Operating system (e.g. Windows XP Professional)

Optional if remote control

- Internet connection with a bandwidth of min. 600 Kbps
- Firewall software
- Internet browser
- Go to my PC Brooker software
- Java Runtime Enviroment (delivered by Bühler AG)

3 Preparation

3.1 Responsibility of Customer

- The complete cabling of the machines and devices which will be integrated into the network.
- Installation of the required software, except the software delivered and installed during the startup by Bühler AG.
- Link to the internet through a firewall (only if remote control)

3.2 Responsibility of Bühler AG

- No special preparation work is needed from Bühler AG, since the full software package is installed on the machine control units by default.

4 Startup

4.1 Data Storage on the Server

4.1.1 Tasks of Customer

Setup a share on the server for the data storage, which is accessible for the connected machine and for the users defined by the customer.

Create a data sheet for the configuration of the network.

Required parameters:

- TCP/IP address of the machine
- TCP/IP address of the server
- TCP/IP address of the gateway
- Subnetmask of the machine network
- Name of the share on the server
- Name of the user assigned to the machine
- Password of the user assigned to the machine

4.1.2 Tasks of Bühler AG

Configuration of the machines connected to the network respectively of the software on the Datanet control unit on the basis of the customer's data sheet.

4.2 Remote control (Office)

4.2.1 Tasks of Customer

Integration into the machine network of one or several Computers, which should have access to the machines.

4.2.2 Tasks of Bühler AG

Together with an IT specialist of the customer, installation and startup of the required software on the computers, connected to the machine network.

4.3 Remote Diagnostics

4.3.1 Tasks of Customer

Setup the computer for remote diagnostics

- connect to the internet through a firewall and with a bandwidth of min. 600Kbps.
- Install an e-mail account used for the remote diagnostics.
- Install the remote control software establishing a secure connection through the internet.

4.3.2 Tasks of Bühler AG

Execute a connection test with all connected machines between the customer service department of Bühler AG and the customer's place.

5 Configuration of the Machine

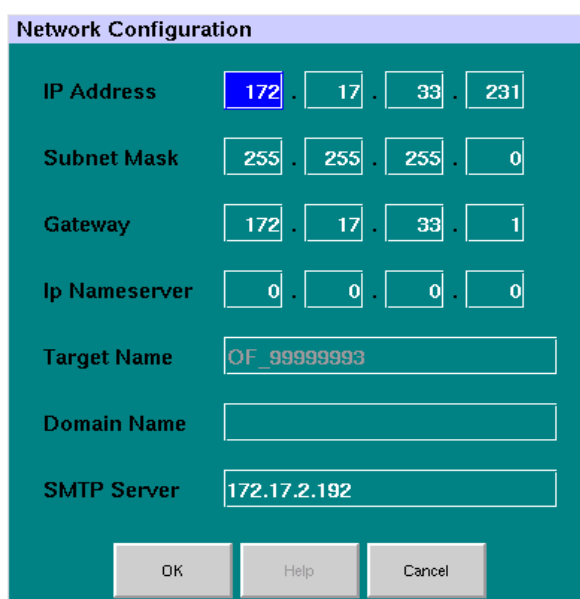
5.1 HMI Settings

Requirements

Logged on to the HMI as a member of the user's group with the access level 4.

Procedure

1.  =>  =>  => 
2. Enter parameters

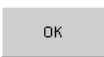


The dialog box titled "Network Configuration" contains the following fields:

- IP Address:** 172 . 17 . 33 . 231
- Subnet Mask:** 255 . 255 . 255 . 0
- Gateway:** 172 . 17 . 33 . 1
- Ip Nameserver:** 0 . 0 . 0 . 0
- Target Name:** OF_99999993
- Domain Name:** (empty)
- SMTP Server:** 172.17.2.192

Buttons at the bottom: OK, Help, Cancel.

- **IP address:**
IP address for the machine. To be defined by the customer.
- **Subnet mask:**
Subnet mask for the machine. To be defined by the customer.
- **Router:**
IP address for the router. To be defined by the customer. (if existing)
- **IP address name server:**
Currently not used.
- **Host name:**
Name of the HMI in the network. To be defined by the customer.
- **Domain name:**
Currently not used.
- **SMTP server:**
Currently not used.

3.  => 

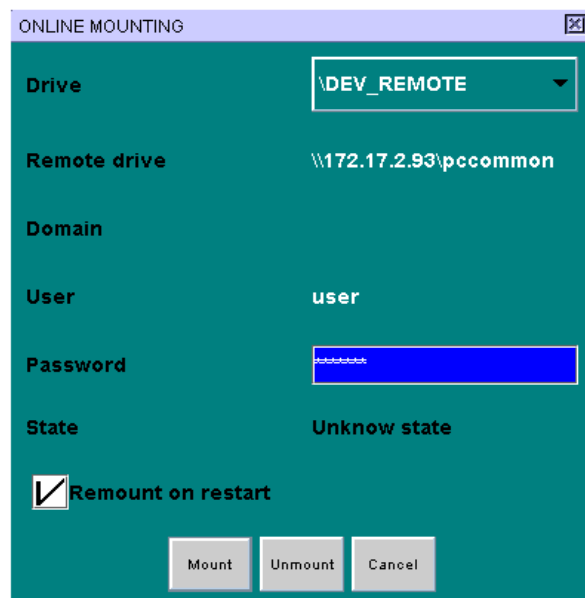
5.2 Customer's Settings

Requirements

Logged on to the HMI as a member of the user's group with the access level 4.

Procedure

1.  =>  =>  => 
2. Enter parameters



- Drive:
DEV_REMOTE, default setting.
- Remote drive:
Path to the server where the machine is connected.
- Domain:
Currently not used.
- User:
User who was setup on the server for the machine.
- Password:
Password for the user who was setup on the server.

3.  => 

6 Software installation

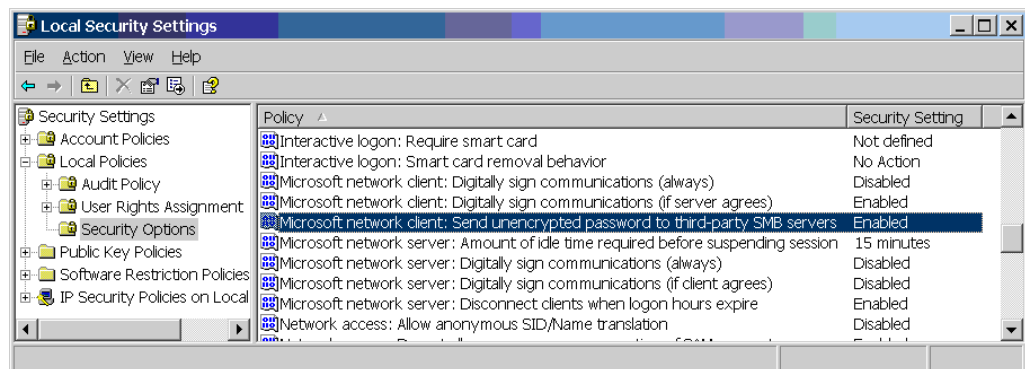
6.1 Jave Runtime Environment

6.1.1 Installation

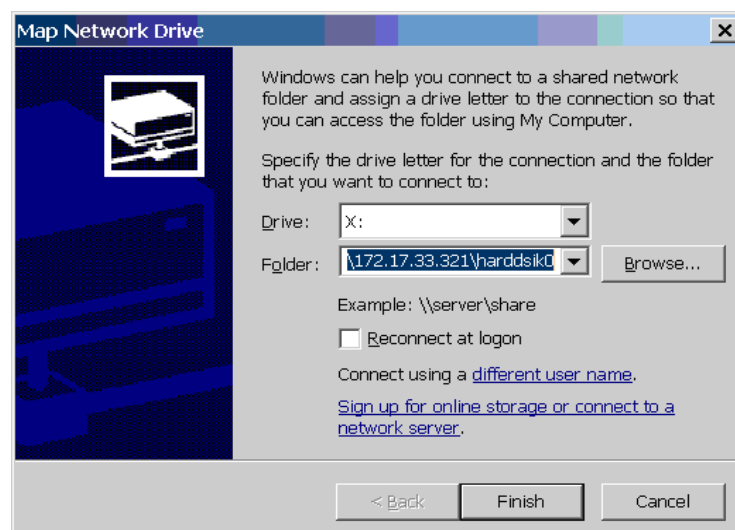
Procedure

1. Check the local safety guideline of the computer.

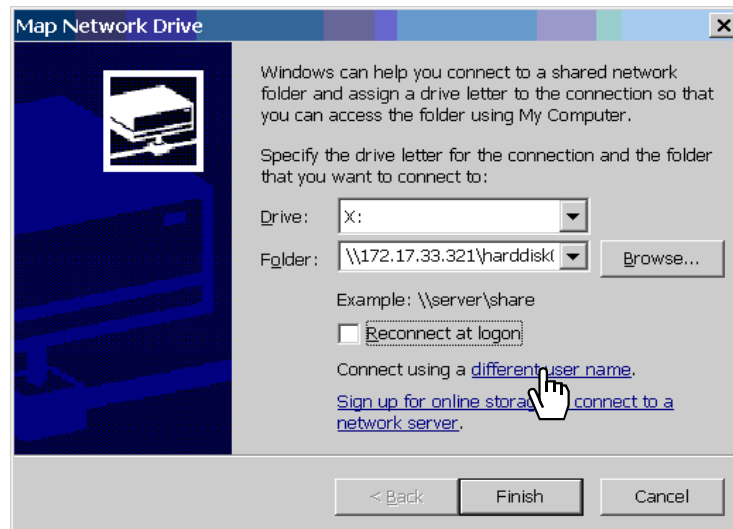
Windows Control Panel => Administrativ Tools => Local Security Settings => Local Policies => Security Options „Microsoft network client:Send unencrypted password to third-party SMB servers“ must be enabled.



2. Windows Explorer => Tools => Map Network Drive.
3. Select a free drive (X:) and enter in the field „Folder“ the [IP-Adress of the machine]\harddisk0 (e.g. \\172.17.10.223\harddisk0).

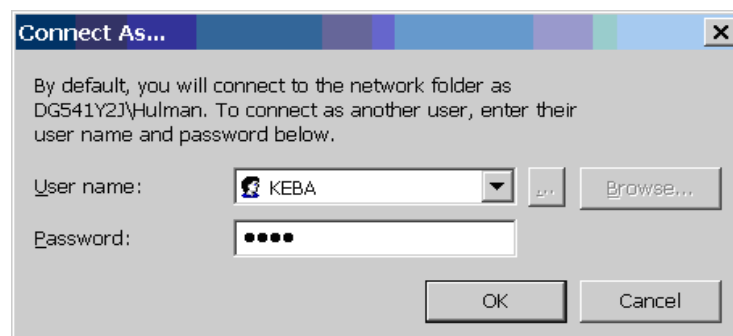


4. Click „Connect adifferent user name“.

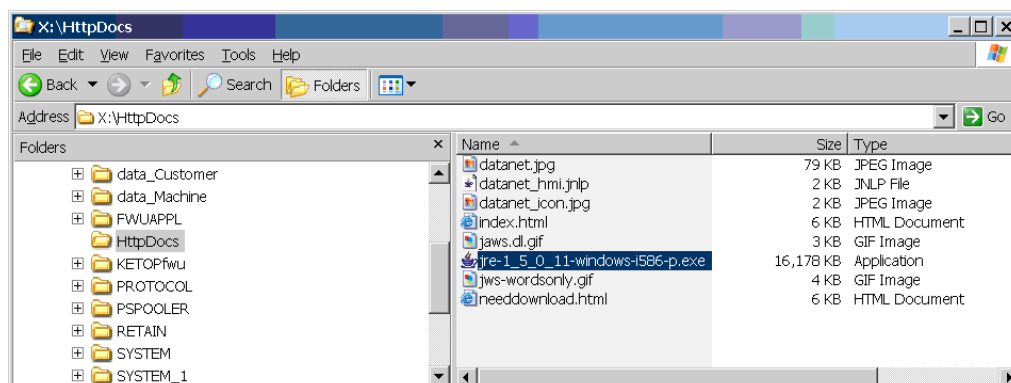


5. Enter the login information.

- User name: KEBA
- Password: KEBA



6. Open the folder „HttpDocs“ and copy the file „jre-*.exe (Java Runtime Enviroment)“ to a temporary directory of the computer.



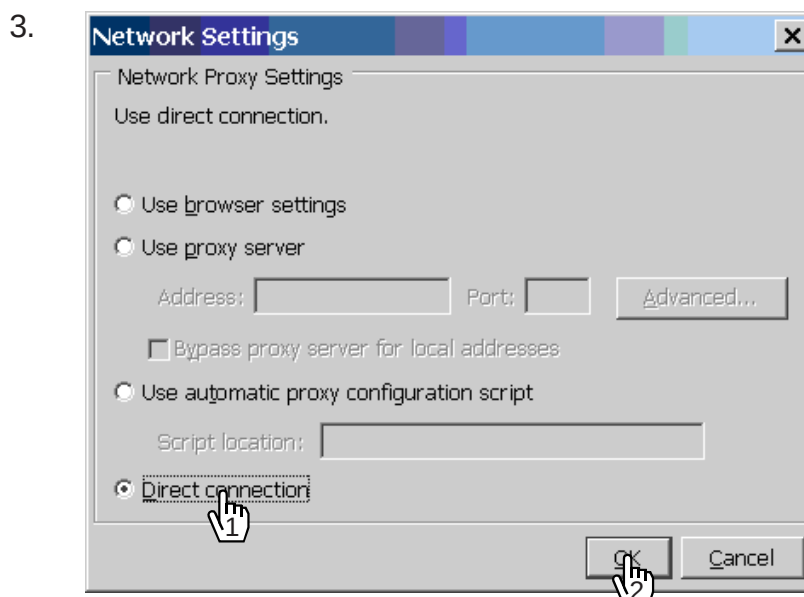
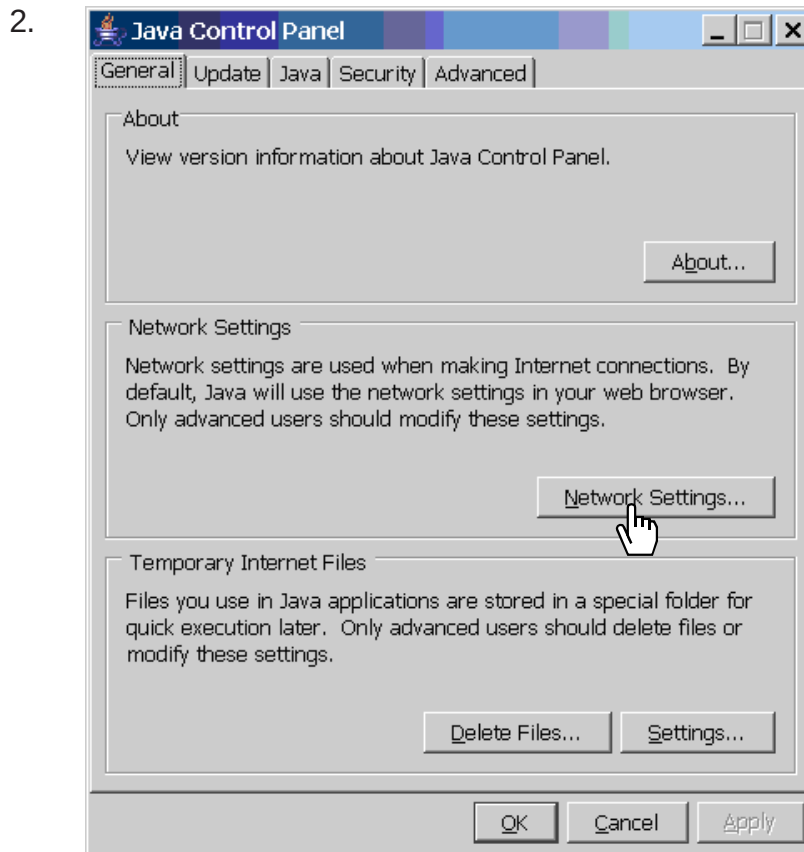
7. Execute the file „jre-*.exe (Java Runtime Enviroment)“. Use the default parameters for the installation.

6.2. Settings

6.2.1 Java Runtime Enviroment without Proxy-Server

Procedure

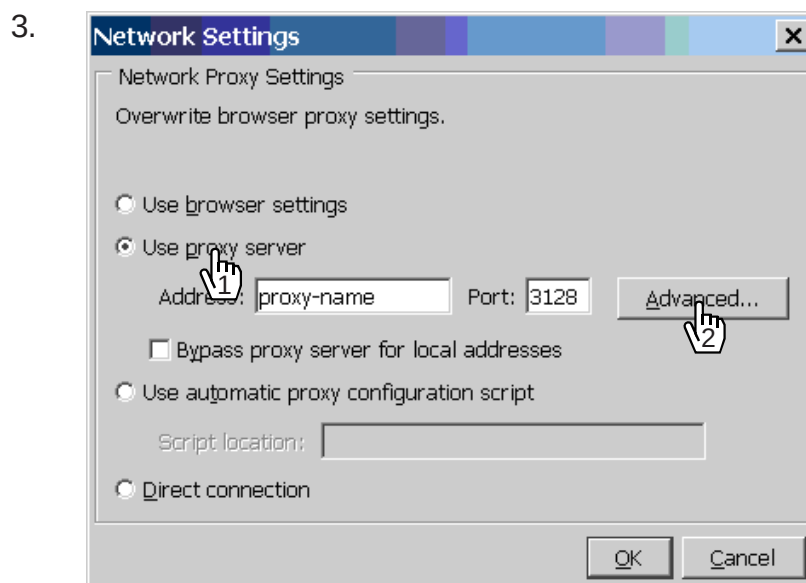
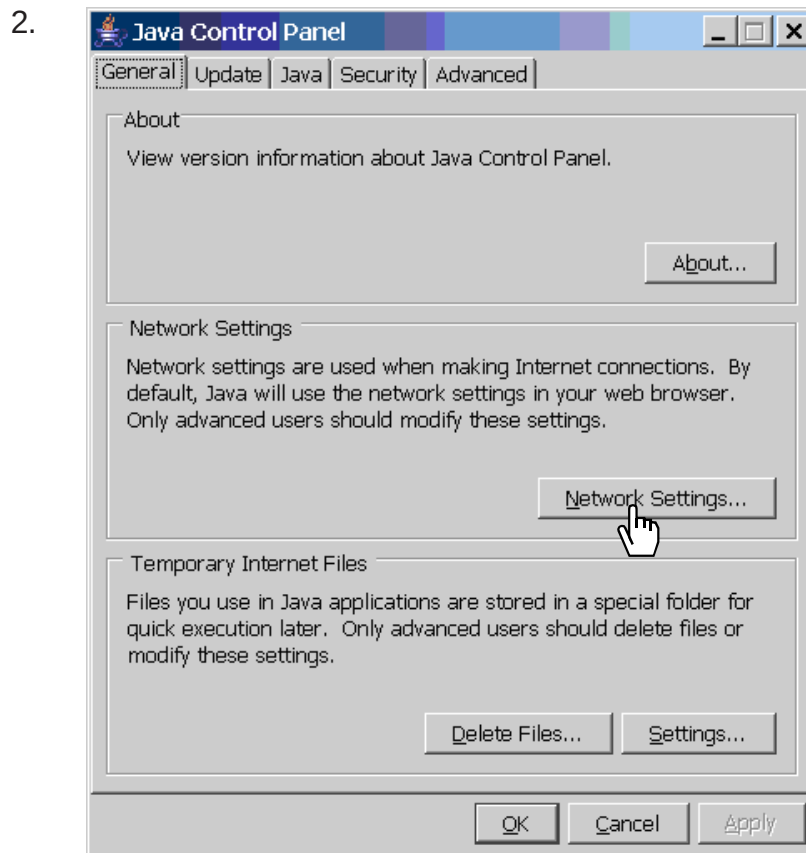
1. Windows control panel => open Java



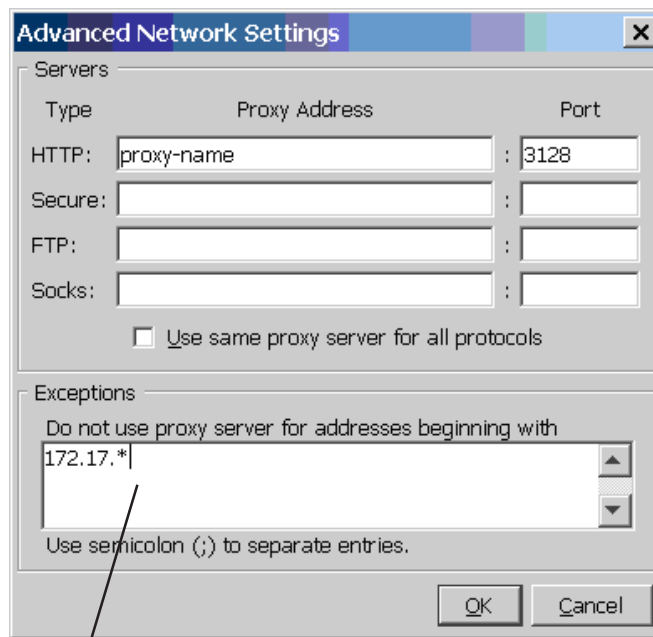
6.2.2 Java Runtime Enviroment with Proxy-Server

Procedure

1. Windows control panel => open Java



4.



The image shows a Windows-style dialog box titled "Advanced Network Settings". It has a close button (X) in the top right corner. The dialog is divided into two main sections: "Servers" and "Exceptions".

The "Servers" section contains a table with three columns: "Type", "Proxy Address", and "Port".

Type	Proxy Address	Port
HTTP:	proxy-name	3128
Secure:		
FTP:		
Socks:		

Below the table is a checkbox labeled "Use same proxy server for all protocols".

The "Exceptions" section contains a text box with the label "Do not use proxy server for addresses beginning with". The text box contains "172.17.*". To the right of the text box are up and down arrow buttons. Below the text box is the instruction "Use semicolon (;) to separate entries."

At the bottom right of the dialog are "OK" and "Cancel" buttons.

Enter the TCP/IP address of the connected machine into this field.

Note



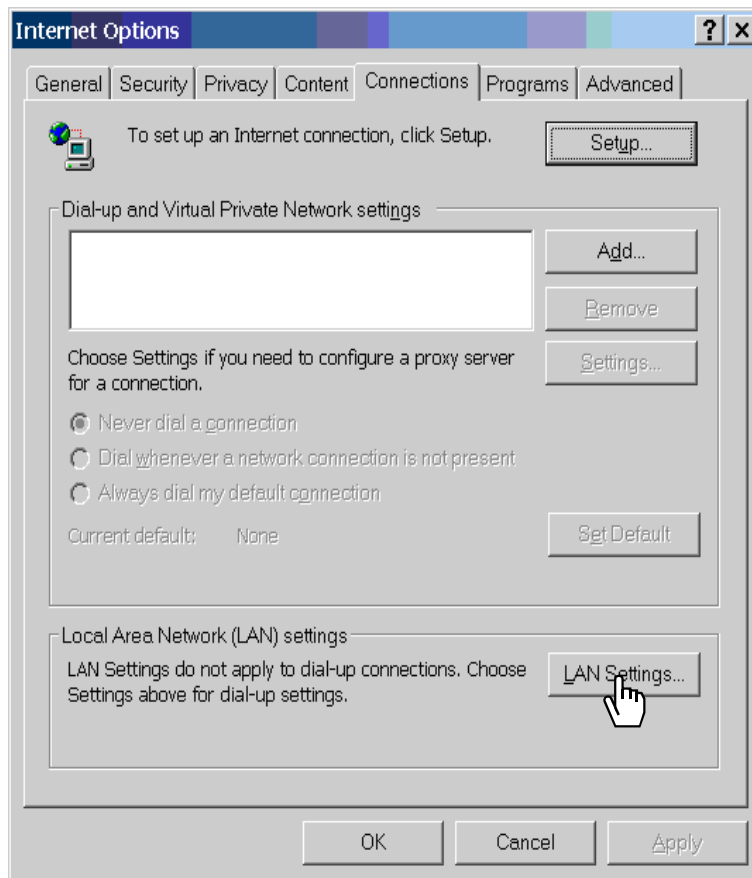
With several machines inside the same subnet, not all addresses must be entered. The changing numbers can be replaced by the wild-card character „star“ () .e.g. 172.17.**

6.2.3 Internet-Explorer without Proxy-Server.

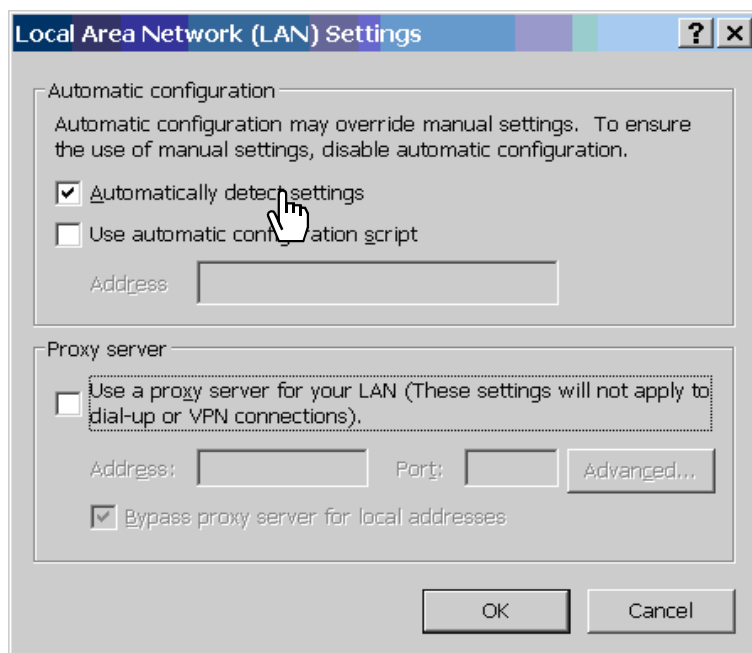
Procedure

1. Open Internet Explorer => Tools => Internet Options => Connections.

- 2.



- 3.

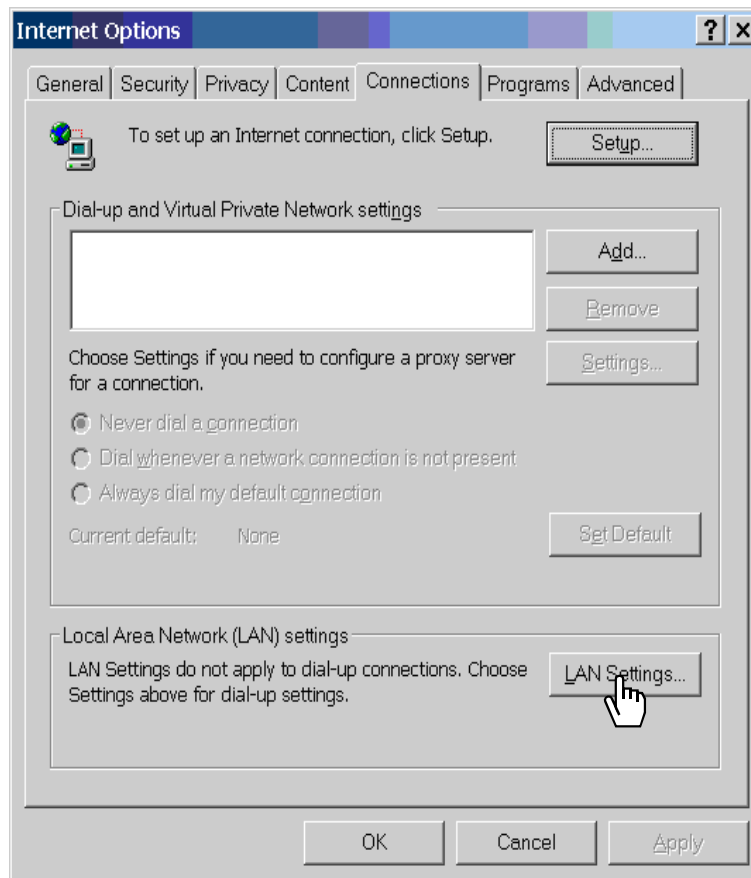


6.2.4 Internet-Explorer without Proxy-Server.

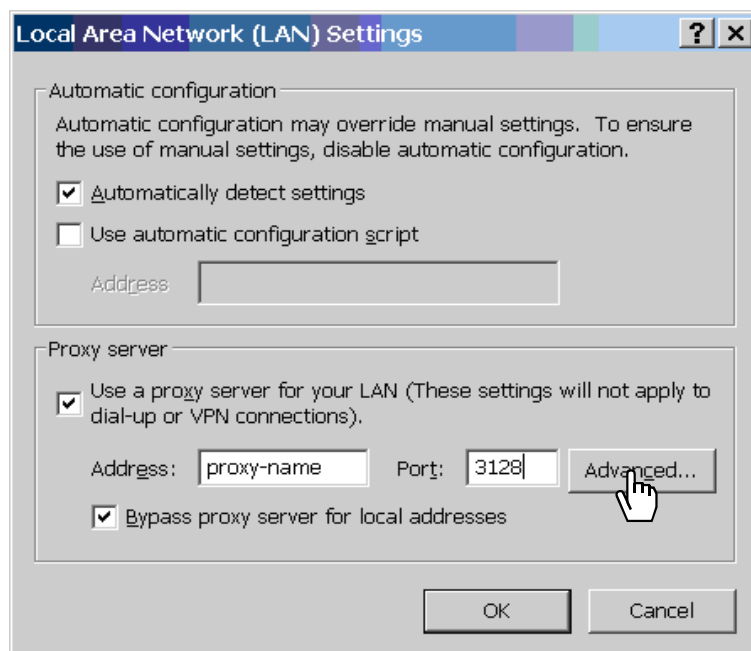
Procedure

1. Open Internet Explorer => Tools => Internet Options => Connections.

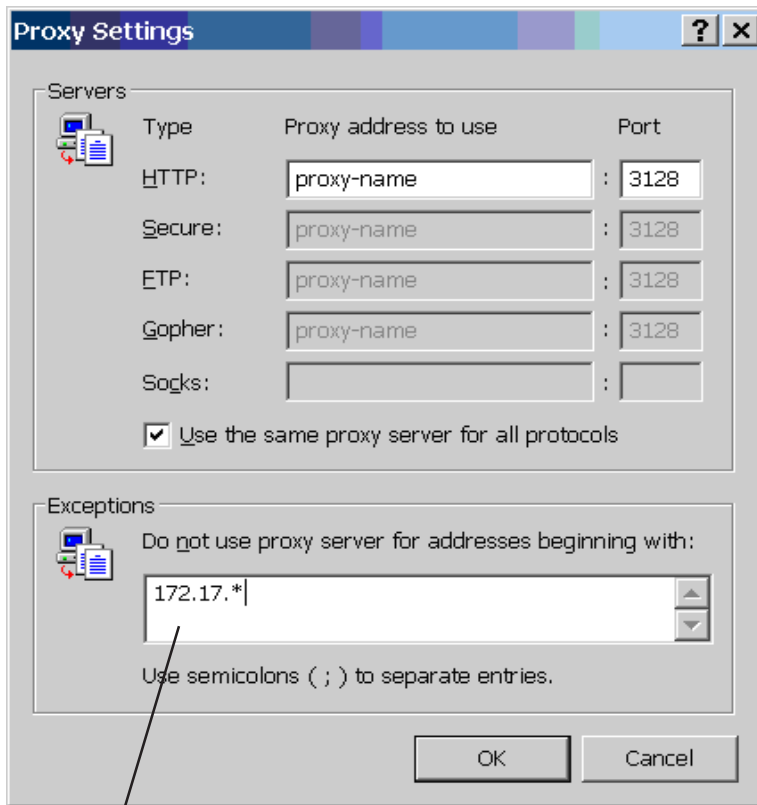
2.



3.



4.



Type	Proxy address to use	Port
HTTP:	proxy-name	3128
Secure:	proxy-name	3128
ETP:	proxy-name	3128
Gopher:	proxy-name	3128
Socks:		

☒ Use the same proxy server for all protocols

Exceptions

Do not use proxy server for addresses beginning with:

172.17.*

Use semicolons (;) to separate entries.

OK Cancel

Enter the TCP/IP address of the connected machine into this field.

Note



With several machines inside the same subnet, not all addresses must be entered. The changing numbers can be replaced by the wild-card character „star“ (*) e.g. 172.17.*

This procedure is only valid for Microxoft Internet Explorer, but the setting must also be done in other browsers.

7 Remote control

7.1 By Internet Browser (e.g. Windows Internet Explorer)

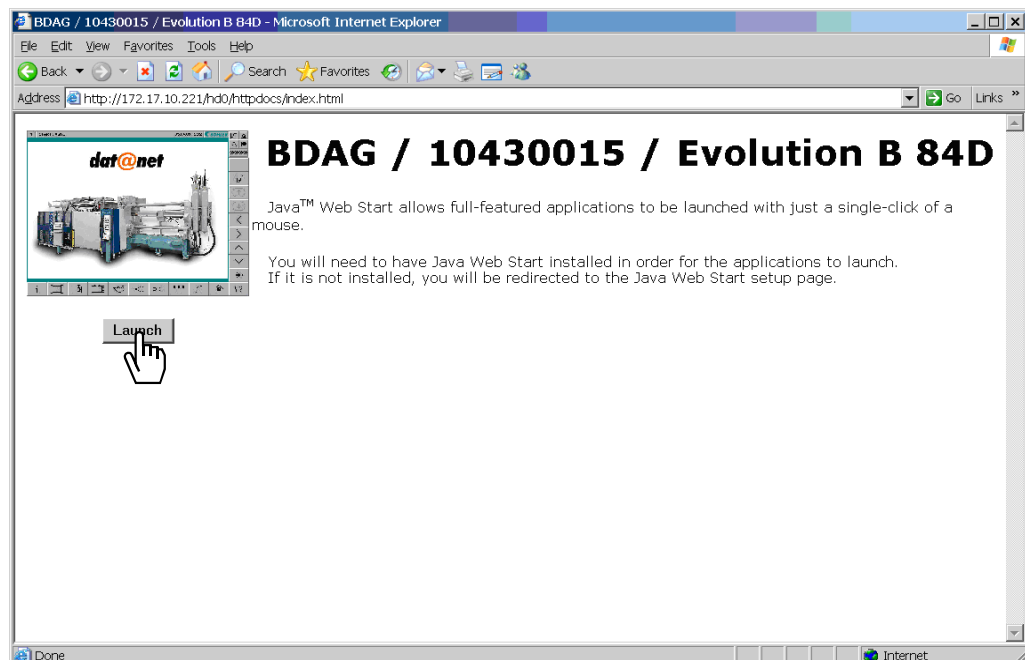
Procedure

1. Open the Internet Explorer.
2. Enter the address <http://172.17.10.221/hd0/httpdocs/index.html>

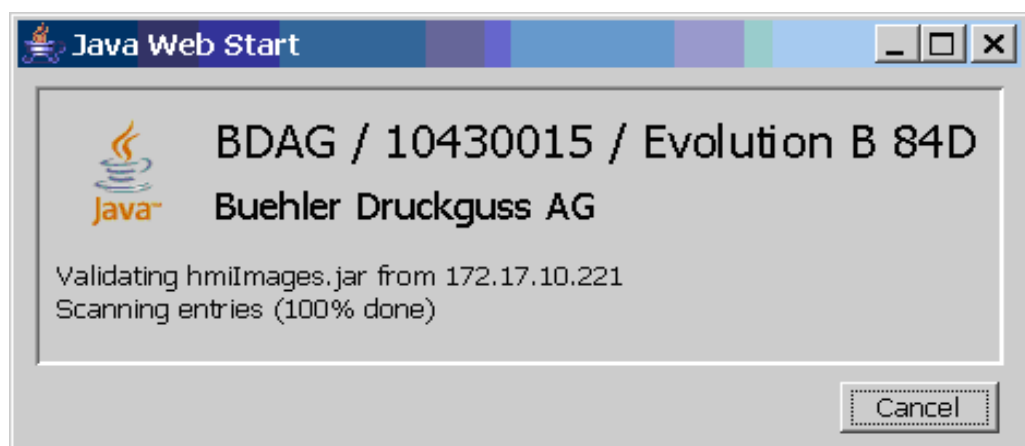
Note



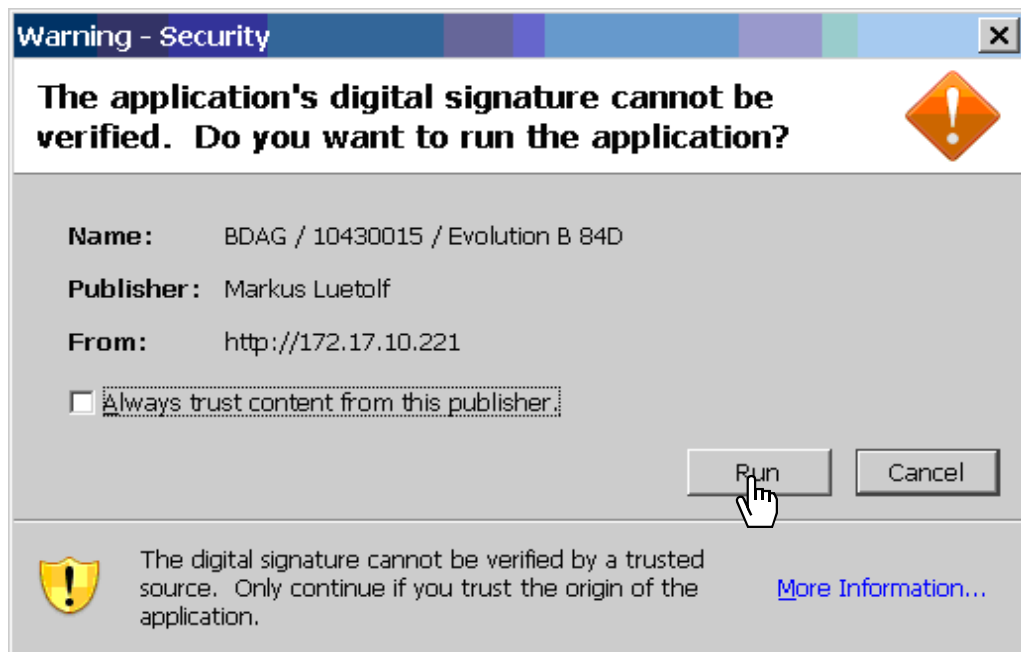
We recommend you to create a favorite in the internet explorer for each machine connected to the network.



Files for the remote control are being copied to the computer.



4.

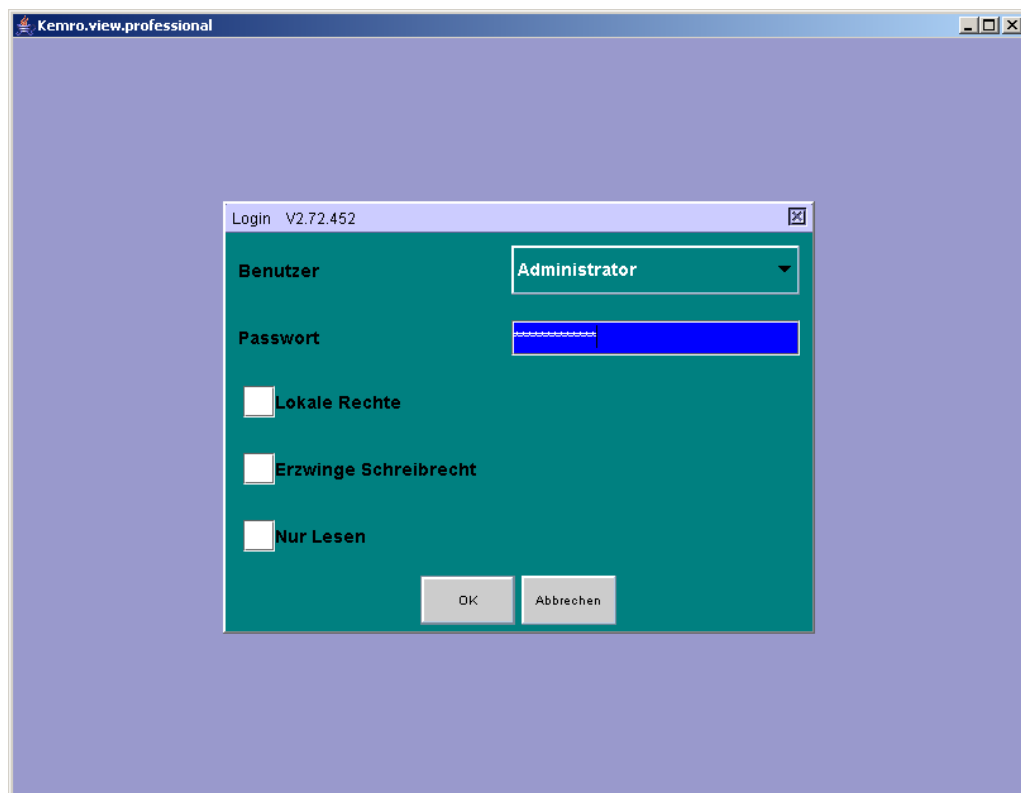


Note



If the application is not executed, access to the machine will not work.

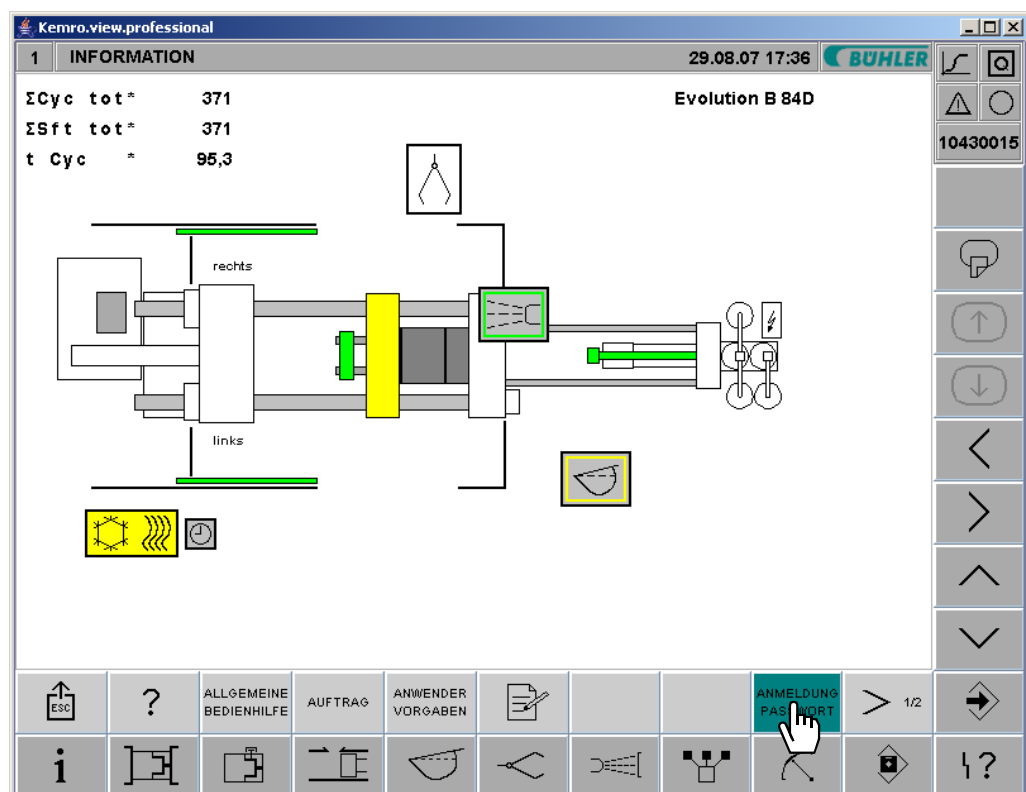
5. Login onto the machine control.



6.



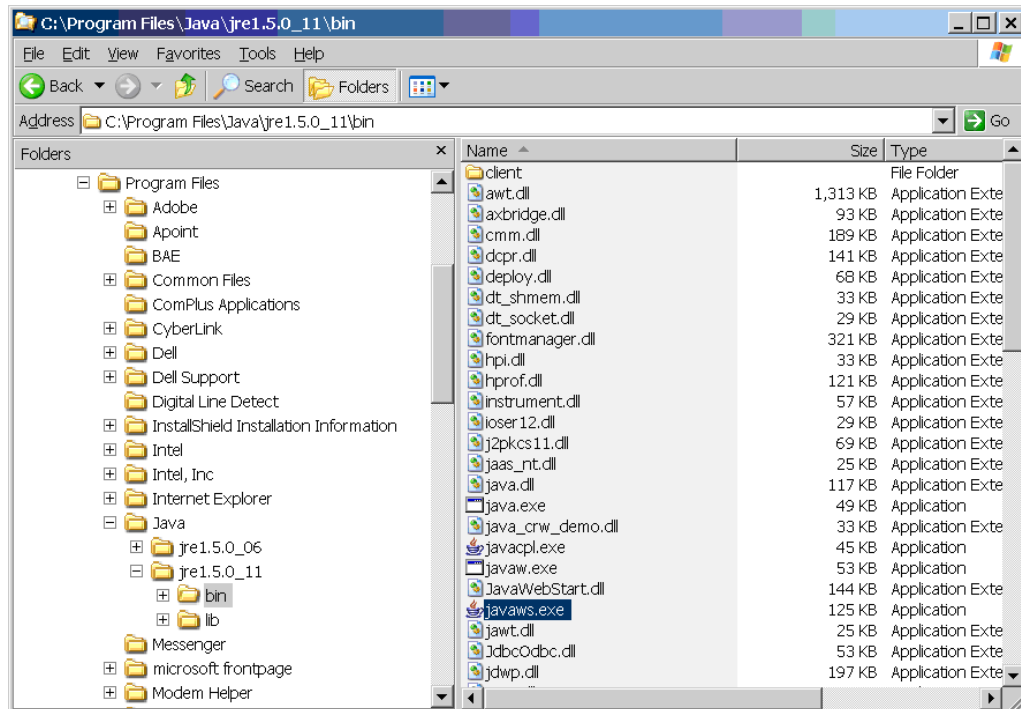
7.



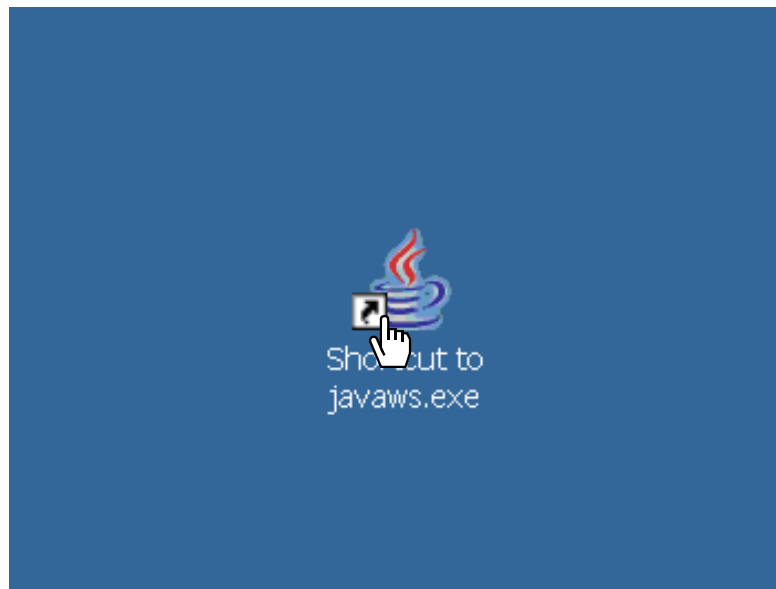
7.2 By JavaWebstart

Procedure

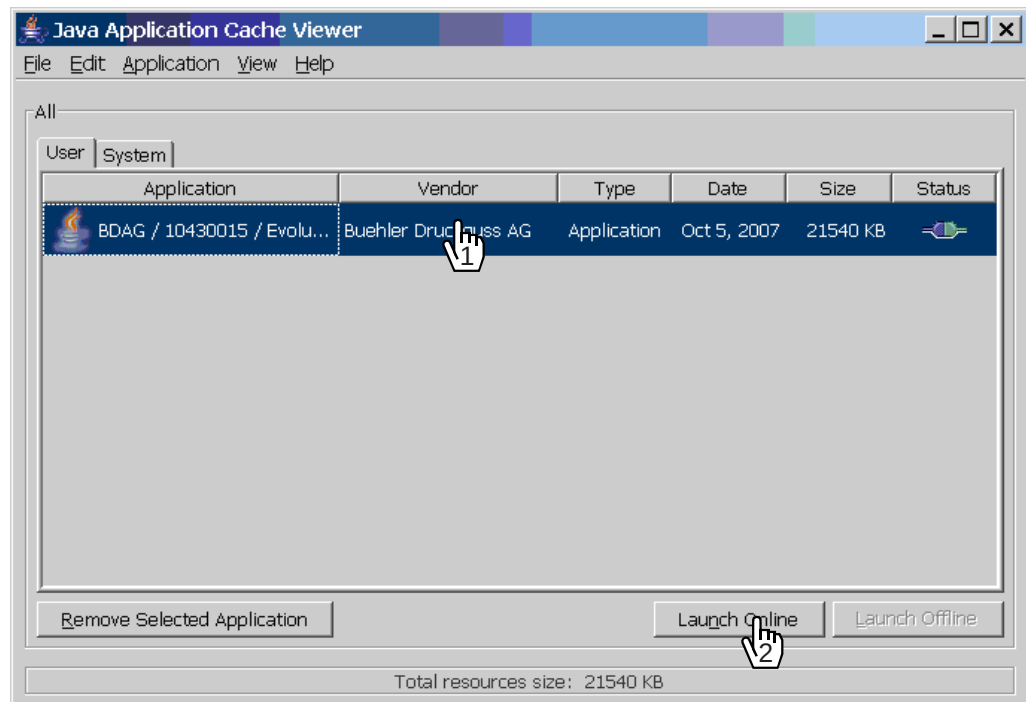
1. Open the Internet Explorer.
2. Create a short cut of the file „javaws.exe“ (Java Webstart) on your desktop.



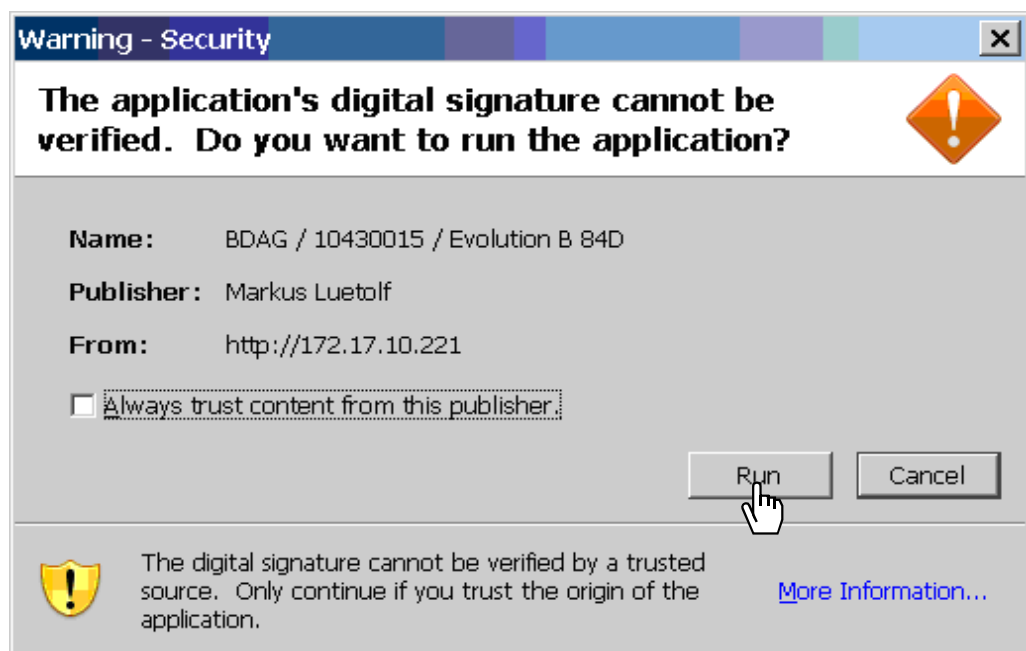
3. Doubleclick the short cut Java Webstart.



4.

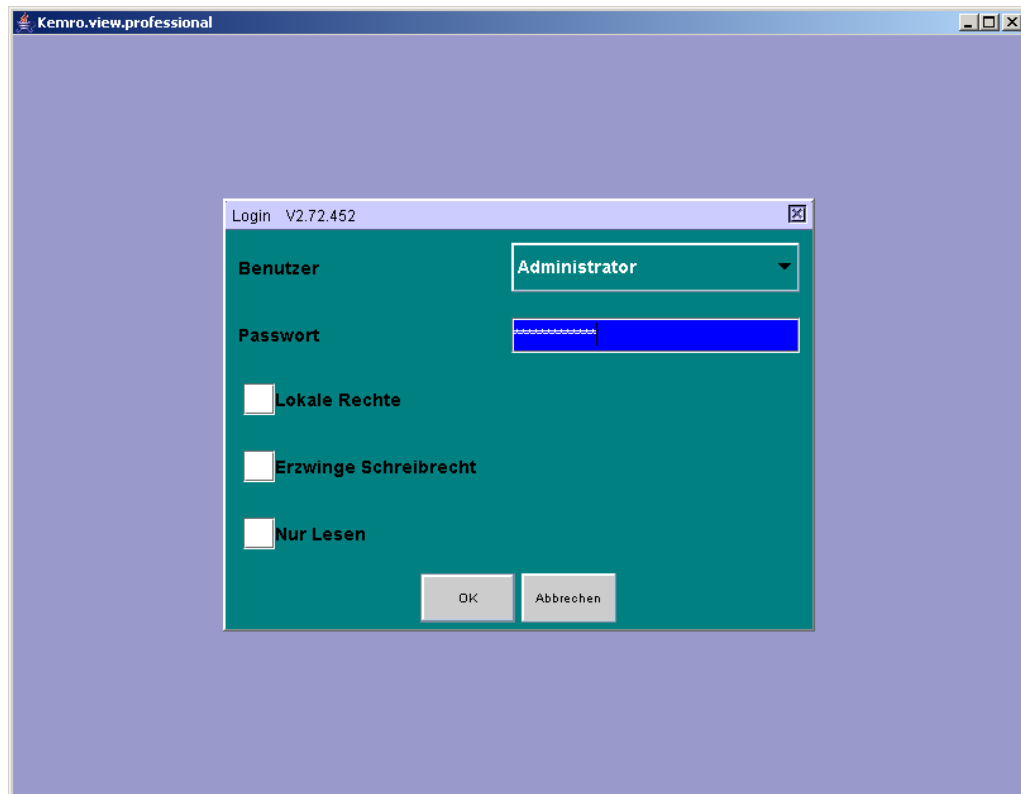


5.

**Note**

If the application is not executed, access to the machine will not work.

6. Login onto the machine control.



- 7.



8.

